Symmetric Tensor Categories (lecture notes)

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About these notes

These notes accompany Guillermo Sanmarco's Winter 2025 course at the University of Washington. This revised edition combines the student-written notes with the instructor's handwritten notes for Lectures 1–23. Statements, notation and proofs have been corrected and expanded. Dates for Lectures 7–9 and 11–23 are estimated from the recorded dates and an assumed Monday/Wednesday/Friday schedule, excluding university holidays. They are marked "est. in the lecture headings. The remaining dates come from the original typed notes.

Scribes

- Wolfgang Allred: Lectures 1 and 7.
- Justin Bloom: Lecture 2.
- Andrew Aguilar: Lecture 3.
- Charlie Magland: Lecture 4.
- Jackson Morris: Lectures 5 and 10.
- Jaedon Rich: Lecture 6.
- Guillermo Sanmarco: Lectures 8, 9 and 11–23.

The February 5 student notes, originally left unnumbered, are Lecture 10. The original Lecture 8 file contained only a heading; its text has been reconstructed from the instructor's notes. Existing student credits are retained for the revised lectures they originally scribed.

Conventions

All tensor products of vector spaces are over the ground field k unless otherwise stated. Representations in a tensor category are finite-dimensional; arbitrary-dimensional comodules are used explicitly when needed for reconstruction. Hypotheses on the field and its characteristic are stated where they enter. A reference accompanying a theorem whose proof is not included identifies the external result being used.

Errata

1. **Lecture 10: the use of Chevalley's theorem.** In the corollary following Chevalley's theorem, use preservation of H-stable lines in representations of the ambient group in place of fixing H-invariant vectors. The reconstruction argument is repaired by applying naturality to the inclusion $L \hookrightarrow W$ of an H-stable line: this gives $\lambda_W(L \otimes R) = L \otimes R$. Chevalley's theorem then identifies $H = G_X$ with the tensor automorphism group. The reduction to G_X and the passage to the inverse limit are unchanged.

2. **Lecture 10: coordinate Hopf algebras.** In the paragraph preceding the inverse limit, "finite-dimensional sub-Hopf algebras" should read "Hopf subalgebras finitely generated as k-algebras."

3. **Lecture 21: the dimension lemma.** For a tensor functor $F : \mathcal{C} \to \mathcal{D}$, the right-hand side of the inequality in the notes should be $\mathrm{FPdim}(\mathcal{C})$:

$$\mathrm{FPdim}(F(\mathcal{C})) \leq \mathrm{FPdim}(\mathcal{C}),$$

   with equality if and only if F is fully faithful. Here $F(\mathcal{C})$ is the full fusion subcategory generated by the image.

1 Lecture 1 (January 6, 2025): Introduction

Scribe: Wolfgang Allred

Introduction to Affine Group Schemes

For the first few lectures we will follow [Wat79]. There are multiple ways to view affine group schemes, but in order to avoid topology (for the time being) we will look at affine group schemes from the "Functor of Points" perspective. Throughout the course, k is an algebraically closed field unless otherwise stated.

Fix equations $f_1, \dots, f_l \in k[x_1, \dots, x_n]$ and put $A = k[x_1, \dots, x_n]/(f_1, \dots, f_l)$. For each commutative unital k-algebra R, their solutions in R^n correspond naturally to algebra maps $A \to R$. Thus they define the functor

$$R \longmapsto \mathrm{Hom}_{k\text{-}\mathsf{Alg}}(A, R).$$

The equations retain more information than their set of k-valued solutions. For example, $x = 0$ and $x^2 = 0$ have the same solutions in k, but different solutions in $k[\epsilon]/(\epsilon^2)$: the second equation also admits $x = \epsilon$. This is why we keep the coordinate algebra, including its nilpotents.

1.1 Definition. Let $\mathcal{C}$ be a locally small category. A functor $F : \mathcal{C} \to \mathsf{Set}$ is **representable** if there exists some $A \in \mathcal{C}$ and a natural isomorphism $F \cong \mathrm{Hom}_{\mathcal{C}}(A, -)$.

An **affine scheme** is a representable functor $X : k\text{-}\mathsf{Alg} \to \mathsf{Set}$. If $X \cong \mathrm{Hom}_{k\text{-}\mathsf{Alg}}(A, -)$ then we say that A is the **coordinate algebra** of X and denote $k[X] \coloneqq A$.

1.2 Example.
1) Affine space $\mathbb{A}^n$ sends $R \mapsto R^n$, so $k[\mathbb{A}^n] = k[x_1, ..., x_n]$.
2) The punctured line $\mathbb{G}_m$ sends $R \to R^*$ where R^* is the set of multiplicative units of R. Here, $k[\mathbb{G}_m] = k[x, y]/(xy - 1)$.

These definitions make sense due to Yoneda's lemma, which is more general than the version that we state below:

1.3 Theorem (Affine-scheme form of Yoneda's lemma)**.** *There is an anti-equivalence between k-*Alg *and* $\mathsf{AffScheme}$ *where $A \mapsto \mathrm{Hom}_{k\text{-}Alg}(A, -)$. In particular, for affine schemes X and Y,*

$$\mathrm{Hom}(X, Y) \cong \mathrm{Hom}_{k\text{-}\mathsf{Alg}}(k[Y], k[X])$$

Proof. A natural transformation $\mathrm{Hom}(A, -) \to \mathrm{Hom}(B, -)$ sends id_A to a map $u : B \to A$. Naturality forces its component at R to send $f : A \to R$ to $f \circ u$. Conversely, composition with any such u defines a natural transformation. This proves full faithfulness; essential surjectivity is the definition of an affine scheme. ∎

1.4 Definition. An **affine group scheme** over k is a functor $G : k\text{-}\mathsf{Alg} \to \mathsf{Group}$ whose underlying set-valued functor is representable.

1.5 Example. 1) $\mathbb{G}_a$ sends $R \mapsto R^+$ where R^+ is the additive group of the underlying ring R. Then $k[\mathbb{G}_a] = k[x]$.

2) $\mathbb{G}_m$ as above.

3) GL_n sends $R \mapsto \mathrm{GL}_n(R)$, the group of invertible $n \times n$ matrices over R. Then $k[\mathrm{GL}_n] = k[x_{ij}, \frac{1}{\det(x_{ij})} \mid 1 \le i, j \le n]$.

4) SL_n sends $R \mapsto \mathrm{SL}_n(R)$ where $\mathrm{SL}_n(R)$ is the group of invertible $n \times n$ matrices with determinant 1 over R. Its coordinate algebra is $k[\mathrm{SL}_n] = k[x_{ij}]/(\det(x_{ij}) - 1)$.

1.6 Definition. A **Hopf algebra** over k is a vector space H with maps

$$m : H \otimes H \to H, \quad u : k \to H, \quad \Delta : H \to H \otimes H, \quad \epsilon : H \to k, \quad S : H \to H$$

satisfying the following axioms.

1. (H, m, u) is an associative unital algebra:

$$m(m \otimes \mathrm{id}) = m(\mathrm{id} \otimes m), \qquad m(u \otimes \mathrm{id}) = \mathrm{id} = m(\mathrm{id} \otimes u).$$

2. (H, Δ, ϵ) is a coalgebra:

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta, \qquad (\epsilon \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \epsilon)\Delta.$$

   Here the usual identifications with $k \otimes H$ and $H \otimes k$ are understood.

3. The maps Δ and ϵ are unital algebra homomorphisms, where $H \otimes H$ has componentwise multiplication.

4. The **antipode** satisfies

$$m(S \otimes \mathrm{id})\Delta = u\epsilon = m(\mathrm{id} \otimes S)\Delta.$$

The first three conditions define a **bialgebra**. The antipode is an anti-algebra map; for a commutative Hopf algebra it is an algebra map.

In the next lecture, the natural multiplication, identity and inverse of an affine group scheme will produce Δ, ϵ and S on its coordinate algebra.

2 Lecture 2 (January 13, 2025): Affine groups and Hopf algebras

Scribe: Justin Bloom

More on affine group schemes

Last time: We defined affine schemes over k as *representable* functors

$$X : k\text{-}\mathsf{Alg} \to \mathsf{Set},$$

represented by a coordinate algebra denoted $k[X]$ (or $\mathcal{O}(X)$). The Yoneda lemma establishes a contravariant equivalence (or antiequivalence) of categories

$$\mathsf{AffSch}_k \underset{\text{Spec}}{\overset{k[\ ]}{\rightleftarrows}} k\text{-}\mathsf{Alg}\ ,$$

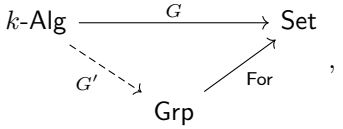
where the functor Spec is defined by $\operatorname{Spec} A : B \mapsto \operatorname{Hom}_{k\text{-}\mathsf{Alg}}(A, B)$.

Today: We will define the category of *affine group schemes* over k, denoted AffGrp_k, as well as the category of *Hopf algebras* over k, denoted Hopf_k. We will see that the antiequivalence above descends to a contravariant equivalence

$$\mathsf{AffGrp}_k \underset{\text{Spec}}{\overset{k[\ ]}{\rightleftarrows}} \mathsf{CommHopf}_k\ ,$$

where $\mathsf{CommHopf}_k$ is the category of commutative Hopf algebras and Hopf-algebra homomorphisms. Its morphisms preserve multiplication, unit, comultiplication, counit and antipode.

2.1 Definition. An *affine group scheme* is an affine scheme G which factors through the category of groups, i.e. such that there exists a functor G' making the diagram commute



,

where For is the forgetful functor taking any group to its underlying set. A *homomorphism* of affine group schemes $G_1 \to G_2$ is a natural map of diagrams, i.e. a natural map such that each $G_1'(R) \to G_2'(R)$ is a group homomorphism for any algebra R.

We will adopt the convention that the forgetful functor is ignored, and an affine group scheme G has target category Grp for simplicity.

2.2 Example. Consider $\mathbf{pt} \in \mathsf{AffSch}_k$, represented by k as an algebra over itself. We have $\mathbf{pt}(R) = \operatorname{Hom}(k, R)$ consists of a single point $\{*\}$, the structure map $k \to R$ included in defining R as an algebra over k, for any R. Since $\{*\}$ can always be given the structure of the trivial group, making any map $\mathbf{pt}(R) \to \mathbf{pt}(R')$ a group homomorphism, we have $\mathbf{pt}$ is an affine group scheme we may call $\mathbf{1}$.

2.3 Definition. Given $X, Y \in \mathsf{AffSch}_k$, we define

$$\begin{aligned} X \times Y : k\text{-}\mathsf{Alg} &\to \mathsf{Set} \\ R &\mapsto X(R) \times Y(R), \end{aligned}$$

called the *fiber product* of X and Y over k.

Notes:

1) We have $X \times Y \in \mathsf{AffSch}_k$, with representing algebra $k[X] \otimes k[Y]$, because it is a fact that $\otimes$ is the coproduct for commutative algebras over k.

2) The product $\times$ is the direct product in AffSch_k, as can be seen by the antiequivalence. In fact, it agrees with the direct product in the big category of Set-valued functors, as can be checked fiberwise.

2.4 Lemma. *If $G \in \mathsf{AffGrp}_k$, then $k[G] \in \mathsf{Hopf}_k$ (the coordinate algebra is a Hopf algebra).*

Proof. Since we have a functor

$$G : k\text{-}\mathsf{Alg} \to \mathsf{Grp},$$

we get maps in AffSch_k

$$G \times G \xrightarrow{\text{mult}} G, \qquad \mathbf{pt} \xrightarrow{\text{unit}} G,$$

$$G \xrightarrow{\text{inverse}} G.$$

The antiequivalence now gives us homomorphisms of algebras

$$k[G] \otimes k[G] \xleftarrow{\Delta} k[G], \qquad k \xleftarrow{\varepsilon} k[G],$$

$$k[G] \xleftarrow{S} k[G].$$

The associative, unit, and inverse axioms all have diagrams, which upon reversing arrows, give the axioms for coassociative, counit, and antipodes, which define a Hopf algebra. ∎

2.5 Theorem. *The Yoneda lemma establishes an antiequivalence*

$$\mathsf{AffGrp}_k \underset{\text{Spec}}{\overset{k[\ ]}{\rightleftarrows}} \mathsf{CommHopf}_k \ .$$

Proof. An immediate corollary of Lemma 2.4 is that the functor $k[\ ]$ takes any group scheme to a commutative Hopf algebra. We must argue for any $A \in \mathsf{CommHopf}_k$, that $\operatorname{Spec} A$ is an affine group scheme and that the relevant compositions are naturally isomorphic to the identity functors.

Let $G = \operatorname{Spec} A$ for a Hopf algebra A, and let $R \in k\text{-}\mathsf{Alg}$.

1) Let $f, g \in G(R)$. We'll define the group product fg as the composition

$$fg : A \xrightarrow{\Delta} A \otimes A \xrightarrow{f \otimes g} R \otimes R \to R,$$

with the rightmost map being multiplication for R, which for commutative R is indeed a homomorphism of algebras. Hence $fg \in G(R)$.

2) The unit in $G(R)$ is $A \xrightarrow{\varepsilon} k \to R$.

3) The inverse of $f \in G(R)$ is defined by

$$f^{-1} : A \xrightarrow{S} A \xrightarrow{f} R.$$

Since A is commutative, S and hence $f \circ S$ are algebra maps. Coassociativity gives associativity of this product, and the counit identities give the unit. In Sweedler notation,

$$\begin{aligned}(ff^{-1})(a) &= \sum f(a_1) f(S(a_2)) \\ &= f\Big(\sum a_1 S(a_2)\Big) = \epsilon(a) 1_R.\end{aligned}$$

The other antipode identity similarly gives $f^{-1} f = \epsilon(-)1_R$. For an algebra map $h : R \to R'$, composition $f \mapsto h \circ f$ preserves this product and the identity. Thus G is a group-valued functor.

What's left to check is that the group structures and Hopf algebra structures are recovered from the constructions we've defined. In other words, given G, we have $A = k[G]$ is a Hopf algebra as constructed in Lemma 2.4, and must verify that our construction for a group scheme $\operatorname{Spec} A$ recovers G. Similarly, starting with A a commutative Hopf algebra, we have now that $G = \operatorname{Spec} A$ is a group scheme, so we must verify that the Hopf algebra construction of 2.4 recovers the given structure on A. We complete this verification in Lecture 3. ■

3 Lecture 3 (January 15, 2025): Affine groups and representations

Scribe: Andrew Aguilar

3.1 Recovering the Hopf structure

We finish the antiequivalence between affine group schemes and commutative Hopf algebras begun in Lecture 2.

3.1 Theorem. *The coordinate-algebra construction gives an antiequivalence between* AffGrp_k *and* $\mathsf{CommHopf}_k$.

Proof. The two constructions of the previous lecture agree with the underlying Yoneda antiequivalence. It remains to check the additional structures. Let A be a commutative Hopf algebra and let $G(R) = \mathrm{Hom}_{k\text{-}\mathsf{Alg}}(A, R)$ with its convolution product. The two algebra maps

$$\iota_l(a) = a \otimes 1, \qquad \iota_r(a) = 1 \otimes a$$

are points of $G(A \otimes A)$. Their product is

$$(\iota_l \iota_r)(a) = \sum (a_1 \otimes 1)(1 \otimes a_2) = \sum a_1 \otimes a_2 = \Delta(a).$$

Thus the group law recovers the original comultiplication. The identity of $G(k)$ is $\epsilon : A \to k$, and the inverse of the universal point $\mathrm{id}_A \in G(A)$ is $S : A \to A$. This recovers the counit and antipode. Conversely, Yoneda identifies the natural group operations on any affine G with precisely these three algebra maps, so they are recovered as well. Finally, a natural map of affine schemes is a group homomorphism exactly when its coordinate-algebra map commutes with these operations. Hence the constructions are inverse on objects and morphisms, up to their natural Yoneda identifications. ∎

For clarity, the multiplication on $A \otimes B$ used above is

$$(a \otimes b)(a' \otimes b') = aa' \otimes bb'.$$

It is the composite of $\mathrm{id}_A \otimes \tau_{B,A} \otimes \mathrm{id}_B$ with $m_A \otimes m_B$, where $\tau_{B,A}(b \otimes a') = a' \otimes b$.

3.2 Example. The additive group $\mathbb{G}_a$ has coordinate algebra $k[x]$, with

$$\Delta(x) = x \otimes 1 + 1 \otimes x, \qquad \epsilon(x) = 0, \qquad S(x) = -x.$$

These formulas express addition, zero and negation.

3.3 Example. For GL_n, put $X = (x_{ij})$ and $d = \det X$. Its coordinate algebra is $A = k[x_{ij}, d^{-1}]$. Matrix multiplication gives

$$\Delta(x_{ij}) = \sum_{l=1}^{n} x_{il} \otimes x_{lj}, \qquad \Delta(d^{-1}) = d^{-1} \otimes d^{-1}.$$

The identity matrix and inverse matrix give

$$\epsilon(x_{ij}) = \delta_{ij}, \qquad S(x_{ij}) = (X^{-1})_{ij}, \qquad \epsilon(d^{-1}) = 1, \quad S(d^{-1}) = d.$$

3.4 Exercise. Check directly that these formulas respect $dd^{-1} = 1$ and satisfy the counit and antipode identities.

3.5 Example. For $\mathbb{G}_m = \mathrm{GL}_1$ the algebra is $k[x, x^{-1}]$, and

$$\Delta(x) = x \otimes x, \qquad \epsilon(x) = 1, \qquad S(x) = x^{-1}.$$

3.2 Rational representations and comodules

Fix a field k, an affine group scheme G and a k-vector space V. Write $V_R = V \otimes_k R$ and $\mathrm{GL}(V)(R) = \mathrm{Aut}_R(V_R)$. When V is finite-dimensional, the functor $R \mapsto V_R$ is an affine scheme represented by $\mathrm{Sym}(V^*)$, and $\mathrm{GL}(V)$ is an affine group scheme isomorphic to $\mathrm{GL}_{\dim V}$ after choosing a basis. For arbitrary V, we use these functors without asserting that they are affine schemes.

3.6 Definition. A *rational G-module* is a vector space V with an action $G(R) \curvearrowright V_R$ by R-linear automorphisms, natural in every commutative k-algebra R. Equivalently, it is a morphism of group-valued functors $G \to \mathrm{GL}(V)$.

3.7 Definition. Let (A, Δ, ϵ) be a coalgebra over k. A *right A-comodule* is a vector space M with a linear map $\rho : M \to M \otimes A$ such that

$$(\rho \otimes \mathrm{id}_A)\rho = (\mathrm{id}_M \otimes \Delta)\rho, \qquad (\mathrm{id}_M \otimes \epsilon)\rho = \mathrm{id}_M .$$

These are coassociativity and counitality. In Sweedler notation we write $\rho(m) = \sum m_0 \otimes m_1$, with the summation understood to be finite for each m.

3.8 Theorem. *Let $A = k[G]$. Giving a rational G-module structure on V is equivalent to giving a right A-comodule structure on V. If V is finite-dimensional, these are also equivalent to a morphism of affine group schemes $G \to \mathrm{GL}(V)$.*

We prove this statement and its compatibility with morphisms in the next lecture.

4 Lecture 4 (January 17, 2025): Representations as comodules

Scribe: Charlie Magland

4.1 The representation–comodule dictionary

Let G be an affine group scheme over a field k, and put $A = k[G]$. We prove Theorem 3.8.

Proof. Given a right coaction $\rho(v) = \sum v_0 \otimes v_1$, define, for $g : A \to R$,

$$r_R(g)(v \otimes a) = \sum v_0 \otimes g(v_1)a.$$

This map is R-linear. Coassociativity and the definition of the product in $G(R)$ give $r_R(g)r_R(h) = r_R(gh)$; counitality gives $r_R(e) = \mathrm{id}$. In particular $r_R(g^{-1})$ is its inverse. Composition of algebra maps $R \to R'$ proves naturality.

Conversely, suppose r_R is such a natural action. Evaluate at the universal point $u = \mathrm{id}_A \in G(A)$ and define

$$\rho(v) = r_A(u)(v \otimes 1).$$

For every $g : A \to R$, naturality gives $r_R(g)(v \otimes 1) = (\mathrm{id}_V \otimes g)\rho(v)$, so this coaction, if valid, recovers the action. Over $A \otimes A$, the two points $\iota_l(a) = a \otimes 1$ and $\iota_r(a) = 1 \otimes a$ multiply to Δ. Applying the action law and naturality to this identity gives $(\rho \otimes \mathrm{id})\rho = (\mathrm{id} \otimes \Delta)\rho$. Evaluating at $\epsilon : A \to k$, the identity of $G(k)$, gives counitality. The displayed formulas show that the constructions are inverse. Notice that $u = \mathrm{id}_A$ is the universal point, not generally the identity point $u_A\epsilon$ of $G(A)$. ∎

4.1 Definition. A morphism of right A-comodules $f : M \to N$ is a linear map satisfying

$$\rho_N f = (f \otimes \mathrm{id}_A)\rho_M.$$

By the formulas above, this is equivalent to $G(R)$-equivariance after extension of scalars to every R.

4.2 Example. The *regular comodule* is $A \xrightarrow{\Delta} A \otimes A$. The *trivial comodule* is $k \to k \otimes A$, $1 \mapsto 1 \otimes 1$; it uses the unit $k \to A$, not the counit $A \to k$.

4.2 The abelian category of comodules

The next constructions require only that A be a coalgebra over k. A subspace $N \subseteq M$ is a *subcomodule* if $\rho_M(N) \subseteq N \otimes A$. The quotient then has the well-defined coaction

$$\overline{\rho}(m + N) = \sum (m_0 + N) \otimes m_1.$$

Coassociativity and counitality follow from those on M.

4.3 Theorem. *The category of right A-comodules is a k-linear abelian category. Kernels, cokernels, images and direct sums are computed on the underlying vector spaces. The same holds for the full subcategory of finite-dimensional comodules, with finite direct sums. The forgetful functors are exact.*

Proof. For a comodule map $f : M \to N$, the identity $(f \otimes \mathrm{id}_A)\rho_M = \rho_N f$ and exactness of tensoring over a field give

$$\rho_M(\ker f) \subseteq \ker(f \otimes \mathrm{id}_A) = \ker(f) \otimes A.$$

The image is also a subcomodule, and the ordinary quotient $N/\operatorname{im} f$ inherits a coaction. These constructions have the kernel and cokernel universal properties because their factorizations are linear maps whose coaction identities descend from that of f. Moreover the vector-space isomorphism $M/\ker f \simeq \operatorname{im} f$ respects coactions.

The direct sum has componentwise coaction. For a finite sum $M_1 \oplus M_2$, the usual inclusions i_j and projections p_j are comodule maps and satisfy

$$p_j i_j = \mathrm{id}_{M_j}, \qquad p_k i_j = 0 \ (k \neq j), \qquad i_1 p_1 + i_2 p_2 = \mathrm{id}_{M_1 \oplus M_2}\,.$$

The zero object and these biproducts make the category additive; comodule maps form linear subspaces of ordinary Hom spaces. The kernel, cokernel and image–coimage statements therefore prove that it is abelian. All these constructions preserve finite-dimensionality when the inputs are finite-dimensional. ∎

4.3 Local finiteness

4.4 Theorem. *Every element of an A-comodule belongs to a finite-dimensional subcomodule. Consequently every comodule is the directed union of its finite-dimensional subcomodules.*

Proof. Write $\rho(m) = \sum_{i=1}^{r} m_i \otimes a_i$ with the a_i linearly independent, and let $N = \operatorname{span}\{m_1, \ldots, m_r\}$. Choose linear functionals $\lambda_j \in A^*$ with $\lambda_j(a_i) = \delta_{ij}$. Coassociativity gives

$$\sum_i \rho(m_i) \otimes a_i = \sum_i m_i \otimes \Delta(a_i).$$

Applying $\mathrm{id}_M \otimes \mathrm{id}_A \otimes \lambda_j$ shows $\rho(m_j) \in N \otimes A$. Counitality gives $m = \sum_i \epsilon(a_i) m_i \in N$. Thus N is the required subcomodule. Finite sums of finite-dimensional subcomodules are finite-dimensional, which proves the directed-union assertion. ∎

4.5 Definition. An affine group scheme is *algebraic* if its coordinate algebra is finitely generated as a k-algebra.

4.6 Corollary. *Every affine group scheme over k is an inverse limit of algebraic affine group schemes.*

Proof. It suffices to show that a commutative Hopf algebra A is the filtered union of its finitely generated Hopf subalgebras. Given $a \in A$, choose a finite-dimensional right subcomodule $V \subseteq A$ containing a. In a basis $v_1, \dots, v_n$ write $\Delta(v_j) = \sum_i v_i \otimes c_{ij}$. The coaction identities give

$$\Delta(c_{ij}) = \sum_l c_{il} \otimes c_{lj}, \qquad \epsilon(c_{ij}) = \delta_{ij}.$$

Applying $\epsilon \otimes \mathrm{id}$ to $\Delta(v_j)$ shows that every v_j is a linear combination of the c_{ij}. Hence the algebra generated by the c_{ij} and $S(c_{ij})$ contains a. It is stable under comultiplication and counit, and also under S, since $S^2 = \mathrm{id}$ in a commutative Hopf algebra (inversion squared is the identity). It is therefore a finitely generated Hopf subalgebra. Applying the same argument to finite sets makes these subalgebras a directed family. Taking spectra reverses the resulting filtered union to an inverse limit. ∎

The subalgebras here are finitely generated as algebras, not necessarily finite-dimensional as vector spaces; $k[x]$ already illustrates the distinction.

4.4 Tensor products

For any bialgebra A, the tensor product of right comodules has coaction

$$\rho_{M\otimes N}(m \otimes n) = \sum m_0 \otimes n_0 \otimes m_1 n_1.$$

Equivalently, apply $\rho_M \otimes \rho_N$, interchange the middle factors using $\tau : A \otimes N \to N \otimes A$, then multiply the two A factors. The bialgebra axioms imply the coaction identities. The unit object is the trivial comodule k, and the ordinary associativity and unit maps are comodule maps. When A is commutative, the flip $M \otimes N \to N \otimes M$ is colinear, since $m_1 n_1 = n_1 m_1$. Thus representations of affine group schemes have a symmetric monoidal structure. For a general Hopf algebra the ordinary flip need not be colinear.

5 Lecture 5 (January 22, 2025): Monoidal categories and braidings

Scribe: Jackson Morris

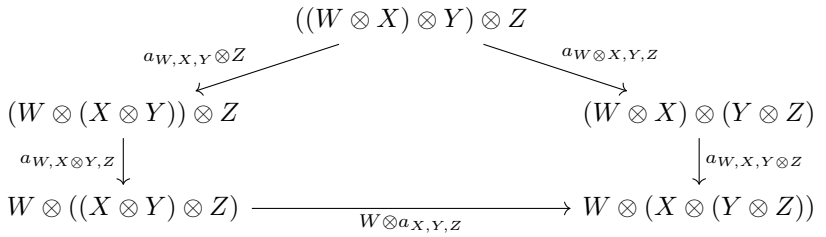
We now introduce monoidal categories and braidings. For further references, see [EGNO15] sections 2, 5, and 8. Last time we defined what an abelian k-linear category was, with the prototypical example being the category G-mod where G is an affine group scheme over k. Today, we abstract those definitions and build on them.

5.1 Definition. A *monoidal structure* on a category $\mathcal{C}$ consists of $(\otimes, a, \mathbb{1}, \iota)$, where:

- $- \otimes - : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ is a bifunctor;
- $a_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$ is a natural isomorphism,

such that the following holds:

1. The following diagram commutes:



   which we call the *pentagon axiom*;

2. There is a pair $(\mathbb{1}, \iota)$ where $\mathbb{1} \in \mathrm{ob}\mathcal{C}$ and

$$\iota : \mathbb{1} \otimes \mathbb{1} \to \mathbb{1}$$

   is an isomorphism such that the functors

$$L_{\mathbb{1}} : X \mapsto \mathbb{1} \otimes X,\ R_{\mathbb{1}} : X \mapsto X \otimes \mathbb{1}$$

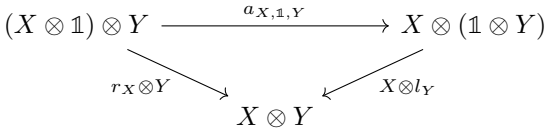
   are equivalences. We call this the *unit axiom*.

5.2 Remark.

- If we are only given a tensor product and associator $(\otimes, a)$ which satisfy the pentagon axiom, then in fact the existence of a pair $(\mathbb{1}, \iota)$ is a property and not any additional structure. More precisely, a unit pair is unique up to a unique isomorphism preserving its multiplication ι.
- Using the functors $L_{\mathbb{1}}$ and $R_{\mathbb{1}}$, we can get natural isomorphisms

$$r : - \otimes \mathbb{1} \to -, \text{ and } l : \mathbb{1} \otimes - \to -$$

for which the *triangle axiom* holds, which is the commutativity of the following diagram:



The following equivalence is proved in [EGNO15, Section 2.2].

5.3 Theorem. *Given a structure $(\mathcal{C}, \otimes, a)$ satisfying the pentagon axiom, then the data of $(\mathbb{1}, \iota)$ satisfying the unit axiom is equivalent to the data $(\mathbb{1}, l, r)$ satisfying the triangle axiom.*

We also use the following coherence theorem; see [EGNO15, Theorem 2.9.2].

5.4 Theorem (**Mac Lane's Coherence Theorem**). *In a monoidal category, given any two parenthesizations P_1 and P_2 of the object $X_1 \otimes \cdots \otimes X_n$, then any two isomorphisms $P_1 \to P_2$ obtained by using associators, their inverses, and tensor products with identity morphisms are actually the same.*

This extends the pentagon axiom to products of arbitrary length and allows us to suppress parentheses in tensor calculations.

Here are some examples of monoidal categories.

5.5 Example. Take Vect_k with the usual tensor product $\otimes_k = \otimes$. Then the associator is the one which "ignores parentheses":

$$a_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z).$$

5.6 Example. Now, take A to be a Hopf algebra and consider the category $A - \mathrm{mod}_k$, where the tensor product is given by tensoring the underlying k-vector spaces, and A acts on $X \otimes_k Y$ by

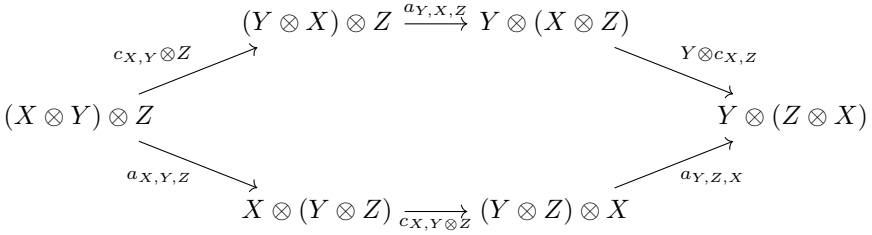
$$a(x \otimes y) = \sum a_1 x \otimes a_2 y,$$

where $\Delta(a) = \sum a_1 \otimes a_2$. The associator is inherited from Vect_k, and the unit is k with action $a \cdot t = \varepsilon(a)t$. Coassociativity gives associativity of the tensor action, while the counit identities give the unit constraints. In fact, a bialgebra suffices for this example.

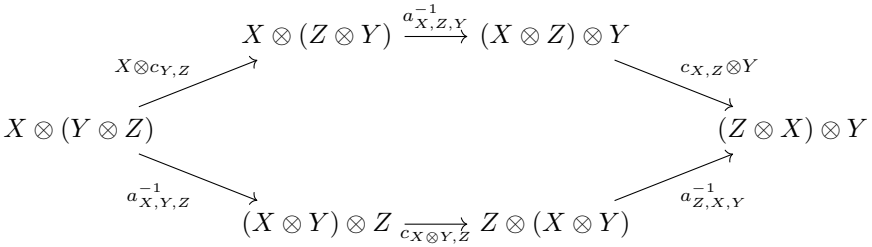
5.7 Definition. A *braided structure* on a monoidal category $(\mathcal{C}, \otimes, a)$ is a natural isomorphism

$$c_{X,Y} : X \otimes Y \to Y \otimes X$$

satisfying two hexagon identities, which are the following commutative diagrams:



and



Suppressing associators, the hexagon identities read

$$c_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes c_{X,Z})(c_{X,Y} \otimes \mathrm{id}_Z),$$

$$c_{X\otimes Y,Z} = (c_{X,Z} \otimes \mathrm{id}_Y)(\mathrm{id}_X \otimes c_{Y,Z}).$$

The associators in the diagrams make these formulas meaningful without a strictness assumption.

5.8 Example. In Vect_k, the braid map is the obvious one:

$$c : X \otimes Y \to Y \otimes X,\ x \otimes y \mapsto y \otimes x.$$

5.9 Example. In $A-\mathrm{mod}_k$, the underlying braiding map of Vect_k lifts to one in $A-\mathrm{mod}_k$ exactly when the flip is A-linear; this holds when $\sum a_1 \otimes a_2 = \sum a_2 \otimes a_1$, namely when A is cocommutative. Conversely, if the flip is A-linear for all modules, apply it to $1\otimes 1$ in the tensor square of the regular module to obtain cocommutativity. For finite-dimensional modules this converse applies when A is finite dimensional. For a particular instance when we can do this, take $A = kG$, the group algebra for some group G. Then for any two G-modules $X, Y \in G-\mathrm{mod}$, then we can place a G-module structure on $X \otimes_k Y$ by

$$g(x \otimes y) = gx \otimes gy,$$

since the comultiplication on $g \in G$ is grouplike by construction. This also is a cocommutative Hopf algebra, and so the underlying braiding from Vect_k lifts.

More generally, a quasitriangular structure on a Hopf algebra gives a braiding on its module category. When A is finite dimensional, all braidings on its finite-dimensional module category arise this way; see [EGNO15, Section 8.3].

6 Lecture 6 (January 24, 2025): Symmetric Braidings and Rigidity

Scribe: Jaedon Rich

6.1 Definition. A braiding $c_{X,Y} : X \otimes Y \to Y \otimes X$ is *symmetric* if

$$c_{Y,X} c_{X,Y} = \mathrm{Id}_{X \otimes Y}.$$

6.2 Remark.

- Let $\mathcal{C}$ be braided and take $X \in \mathcal{C}$. Then we get an action of the braid group

$$B_n = \langle b_1, \ldots, b_{n-1} \mid \begin{array}{ll} b_i b_j = b_j b_i & (|i-j|>1), \\ b_i b_{i+1} b_i = b_{i+1} b_i b_{i+1} & (1 \le i \le n-2) \end{array} \rangle$$

  on $X^{\otimes n}$, with b_i acting by $c_{X,X}$ in positions $i, i+1$ and identities elsewhere. Coherence supplies the required associators.

- If the braiding is symmetric, this action factors through $S_n = B_n / \langle b_i^2 \rangle$

6.3 Example. The "swap" braiding in $\mathbb{V}\mathrm{ec}_k$

$$x \otimes y \mapsto y \otimes x$$

is symmetric.

6.4 Example. For every affine group scheme G, the ordinary flip is G-equivariant for the diagonal action, so its representation category is symmetric as well.

6.5 Example. Take an abstract group G. Define the category $\mathbb{V}\mathrm{ec}_k\, G$ of G-graded k-vector spaces whose objects are vector spaces of the form

$$V = \bigoplus_{g \in G} V_g$$

and whose morphisms are degree-preserving linear maps.
For any G, this has a monoidal structure given by

$$(V \otimes W)_g = \bigoplus_{xy=g} (V_x \otimes W_y).$$

In this case, a braided structure exists if and only if G is abelian. To see the only if direction, notice that for any $g \in G$ we have an object δ_g which is k in degree g and 0 in all other degrees. If G has a braiding, then we get

$$\delta_{gh} = \delta_g \otimes \delta_h \simeq \delta_h \otimes \delta_g = \delta_{hg},$$

and so $gh = hg$. Conversely, for abelian G, the ordinary flip preserves the grading and satisfies the hexagon identities for the usual associator.

6.6 Example. Consider $\mathrm{SVec}_k = \mathbb{V}\mathrm{ec}_k(\mathbb{Z}/2\mathbb{Z})$, vector spaces graded by $\mathbb{Z}/2\mathbb{Z}$. In particular, every object is of the form $V = V_0 \oplus V_1$, where we call V_0 and V_1 the even and odd homogeneous components, respectively.
For $v \in V_i$, we define $|v| = \deg v = i$. Then the braiding $V \otimes W \to W \otimes V$ is given on homogeneous elements by

$$v \otimes w \mapsto (-1)^{|v||w|} w \otimes v.$$

The sign is multiplicative in each degree, so the hexagon identities hold; applying the flip twice gives sign $(-1)^{2|v||w|} = 1$, hence symmetry. When $\operatorname{char} k = 2$, the sign equals 1 and this braiding is the ordinary flip on graded spaces. The distinction considered below requires characteristic different from 2.

6.7 Definition. Fix a symmetric monoidal category $\mathcal{C}$ and $X \in \mathcal{C}$. A *dual object* of X is an object X^* with maps $\mathrm{ev}_X : X^* \otimes X \to \mathbb{1}$ and $\mathrm{coev}_X : \mathbb{1} \to X \otimes X^*$ such that

$$(\mathrm{id}_X \otimes \mathrm{ev}_X) a_{X,X^*,X} (\mathrm{coev}_X \otimes \mathrm{id}_X) = \mathrm{id}_X,$$
$$(\mathrm{ev}_X \otimes \mathrm{id}_{X^*}) a^{-1}_{X^*,X,X^*} (\mathrm{id}_{X^*} \otimes \mathrm{coev}_X) = \mathrm{id}_{X^*},$$

where the canonical unit constraints identify the sources and targets with X and X^*.

6.8 Definition. $\mathcal{C}$ is *rigid* if all objects admit a dual.

6.9 Remark.

- If X^* exists, it is unique up to a unique isomorphism compatible with evaluation and coevaluation.
- Rigidity is a property, not a structure.
- If $\mathcal{C}$ is rigid, we get a contravariant functor $X \mapsto X^*$.
- In a symmetric category, we get a natural isomorphism $X \cong (X^*)^*$.
- Rigidity can be defined for any monoidal category. Without the symmetric braiding, we get "left" and "right" duals X^* and ${}^*X$, respectively, such that $X \cong {}^*(X^*)$.

6.10 Example. In $\mathbb{V}\mathrm{ec}_k$, we can let $V^* \coloneqq \mathrm{Hom}_k(V, k)$ with

$$\begin{aligned} \mathrm{ev}_V : V^* \otimes V &\to k \\ f \otimes v &\mapsto f(v) \end{aligned}$$

and

$$\begin{aligned} \mathrm{coev}_V : k &\to V \otimes V^* \\ 1 &\mapsto \sum_i v_i \otimes f_i, \end{aligned}$$

where $\{v_i\}$ is a finite basis and $\{f_i\}$ its dual basis. A coevaluation satisfying the triangle identities exists precisely when V is finite dimensional. The tensor

$\sum_i v_i \otimes f_i$ corresponds to id_V under $V \otimes V^* \simeq \mathrm{End}_k(V)$, so it is independent of the basis. Conversely, if a coevaluation satisfying the triangle identities has $\mathrm{coev}(1) = \sum_{i=1}^m v_i \otimes f_i$, the first identity gives $v = \sum_i f_i(v) v_i$ for every $v \in V$. Thus the v_i span and V is finite dimensional.

6.11 Definition. An abelian k-linear category $\mathcal{C}$ is *Artinian* if

1) Hom sets are finite dimensional k-vector spaces
2) All objects have finite length. That is, for any $X \in \mathcal{C}$ there is a family of strict inclusions
$$0 = X_0 \subset X_1 \subset \cdots \subset X_n = X$$
such that each X_i/X_{i-1} is simple.

6.12 Definition. A *symmetric tensor category* is a category $\mathcal{C}$ which has

1) A rigid symmetric monoidal structure.
2) An Artinian abelian k-linear structure.

such that

1) $\otimes$ is k-linear on Homs (that is, bilinear).
2) $\mathrm{End}_{\mathcal{C}}(\mathbb{1}) \cong k$. Under these hypotheses the unit is simple; see [EGNO15, Section 4.3].

6.13 Remark. Rigidity implies that tensoring with an object has both a left and a right adjoint, given by tensoring with suitable duals. A functor between abelian categories with both adjoints is exact. Hence $\otimes$ is exact in each variable; bilinearity alone would not imply this. See [EGNO15, Proposition 4.2.1].

6.14 Exercise. Show that G-graded vector spaces are equivalent to right comodules over the group algebra kG, by sending $v \in V_g$ to $v \otimes g$.

7 Lecture 7 (January 29, 2025, est.): Supergroups and categorical trace

Scribe: Wolfgang Allred

A symmetric tensor category is a locally finite k-linear abelian category with a rigid symmetric monoidal structure, bilinear tensor product and $\mathrm{End}(\mathbb{1}) = k$. Examples include finite-dimensional vector spaces and finite-dimensional representations of an affine group scheme. We now explain the analogous super examples. In the discussion of supergroups in this lecture, assume $\mathrm{char}\, k \neq 2$.

Affine supergroups

A *superalgebra* is a $\mathbb{Z}/2\mathbb{Z}$-graded algebra $A = A_0 \oplus A_1$ with $A_i A_j \subseteq A_{i+j}$. It is *supercommutative* when

$$ab = (-1)^{|a||b|} ba$$

for homogeneous elements. In particular, odd elements have square zero. Morphisms preserve the grading. The tensor product of superalgebras has multiplication

$$(a \otimes b)(a' \otimes b') = (-1)^{|b||a'|} aa' \otimes bb'.$$

These are precisely the commutative algebra objects in the symmetric category of all supervector spaces. Although representations will be finite dimensional, the algebras used as coordinate algebras and test algebras need not be.

7.1 Definition. An *affine supergroup scheme* is a group-valued functor on commutative superalgebras whose underlying set-valued functor is represented by a commutative superalgebra $\mathcal{O}(G)$. Equivalently, $\mathcal{O}(G)$ is a commutative Hopf algebra in supervector spaces, with all structure maps even and tensor products using the sign rule above.

As for ordinary affine groups, Yoneda gives a contravariant equivalence between affine supergroups and commutative Hopf superalgebras.

A finite-dimensional representation of G can be described as a supervector space V with an even coaction

$$\rho_V : V \longrightarrow V \otimes \mathcal{O}(G)$$

satisfying the counit and coassociativity identities. Equivalently, it is a natural action of $G(R)$ on $V \otimes R$ by even R-linear automorphisms for every commutative superalgebra R. The correspondence is obtained by evaluating the universal point, exactly as in the ordinary case. Kernels and cokernels are graded kernels and cokernels, the tensor product uses the Hopf multiplication with its signs, and dual representations use the antipode. Thus the category $\mathrm{Rep}_{\mathrm{sVec}}(G)$, with even intertwiners, is a symmetric tensor category.

7.2 Definition. Suppose that $z \in G(k)$ satisfies $z^2 = 1$ and its conjugation action on G is the parity automorphism. Define $\mathrm{Rep}(G, z)$ to be the full subcategory of $\mathrm{Rep}_{\mathrm{sVec}}(G)$ consisting of representations on which z acts by parity:

$$z|_{V_0} = \mathrm{id}_{V_0}, \qquad z|_{V_1} = -\mathrm{id}_{V_1}.$$

This subcategory is closed under finite sums, subquotients, tensor products and duals, and contains the unit; hence it is a symmetric tensor category. Notice that z is central on the even subgroup, but need not be central in the whole supergroup: its conjugation acts by -1 on odd coordinates.

There are two useful special cases.

1. If G is an ordinary affine group scheme, then $\mathrm{Rep}(G, 1)$ consists of purely even representations and is the usual category $\mathrm{Rep}_k(G)$.
2. For any affine supergroup G, let $\widetilde{G} = G \rtimes \mathbb{Z}/2\mathbb{Z}$, where the nontrivial element z acts on G by parity. Every superrepresentation of G extends uniquely to a representation of $\widetilde{G}$ in which z acts by parity. Restriction therefore gives

$$\mathrm{Rep}(\widetilde{G}, z) \simeq \mathrm{Rep}_{\mathrm{sVec}}(G).$$

The parity condition and this construction are discussed in [EGNO15, Section 9.11].

Trace and dimension

7.3 Definition. For an endomorphism $f : X \to X$ of a dualizable object in a symmetric monoidal category, define

$$\mathrm{Tr}(f) : \mathbb{1} \xrightarrow{\mathrm{coev}_X} X \otimes X^* \xrightarrow{f\otimes \mathrm{id}} X \otimes X^* \xrightarrow{c_{X,X^*}} X^* \otimes X \xrightarrow{\mathrm{ev}_X} \mathbb{1}.$$

The *categorical dimension* is $\dim X = \mathrm{Tr}(\mathrm{id}_X)$. These are elements of $\mathrm{End}(\mathbb{1})$, identified with k in a tensor category.

In vector spaces, choose a basis v_i with dual basis v_i^*. The composite sends

$$1 \longmapsto \sum_i v_i \otimes v_i^* \longmapsto \sum_i f(v_i) \otimes v_i^* \longmapsto \sum_i v_i^* \otimes f(v_i) \longmapsto \sum_i v_i^*(f(v_i)),$$

so it is the ordinary trace. In supervector spaces, use a homogeneous basis. The flip of an odd vector past its odd dual contributes a minus sign. For an even endomorphism f,

$$\mathrm{Tr}(f) = \mathrm{tr}(f|_{V_0}) - \mathrm{tr}(f|_{V_1}), \qquad \dim V = \dim_k V_0 - \dim_k V_1.$$

The difference on the right is interpreted in the field k.

For the odd line $\Pi = k^{0|1}$ we have $\Pi \otimes \Pi \simeq \mathbb{1}$, but $\dim \Pi = -1$. In an ordinary representation category, an object whose tensor square is the unit has one-dimensional underlying space, hence categorical dimension 1. The next lecture explains why a symmetric tensor functor preserves these quantities and therefore why sVec cannot be an ordinary representation category when $\mathrm{char}\, k \neq 2$.

8 Lecture 8 (January 31, 2025, est.): Tensor functors and moderate growth

Scribe: Guillermo Sanmarco

The parity element

We first make the condition on the element z from Lecture 7 explicit. Put $A = \mathcal{O}(G)$ and view $z \in G(k)$ as an even algebra homomorphism $z : A \to k$. The pullback of conjugation by z is

$$a \longmapsto \sum z(a_{(1)})\, a_{(2)}\, z^{-1}(a_{(3)}).$$

Here $z^{-1} = z \circ S$, and $z^2 = 1$ makes $z^{-1} = z$. The parity condition requires this expression to equal $a_0 - a_1$ for $a = a_0 + a_1$, $a_i \in A_i$. It does not require z to be central in the whole supergroup.

Monoidal and symmetric tensor functors

8.1 Definition. A *strong monoidal functor* $(F, F_0, F_2) : \mathcal{C} \to \mathcal{D}$ consists of a functor, an isomorphism $F_0 : \mathbb{1}_{\mathcal{D}} \to F(\mathbb{1}_{\mathcal{C}})$, and natural isomorphisms

$$F_{2,X,Y} : F(X) \otimes F(Y) \longrightarrow F(X \otimes Y).$$

They satisfy the associativity identity

$$F(a_{X,Y,Z})F_{2,X\otimes Y,Z}(F_{2,X,Y} \otimes \mathrm{id}) = F_{2,X,Y\otimes Z}(\mathrm{id} \otimes F_{2,Y,Z}) \circ a_{F(X),F(Y),F(Z)},$$

and the two unit identities

$$F(l_X)F_{2,\mathbb{1},X}(F_0 \otimes \mathrm{id}) = l_{F(X)},$$
$$F(r_X)F_{2,X,\mathbb{1}}(\mathrm{id} \otimes F_0) = r_{F(X)}.$$

For braided categories, F is *braided* if

$$F(c^{\mathcal{C}}_{X,Y})F_{2,X,Y} = F_{2,Y,X}c^{\mathcal{D}}_{F(X),F(Y)}.$$

If both braidings are symmetric, we call it a symmetric monoidal functor.

In these notes, a *symmetric tensor functor* between symmetric tensor categories means a k-linear exact strong symmetric monoidal functor. Such a functor is faithful.

Proof of faithfulness. If $X \neq 0$, the evaluation $X^* \otimes X \to \mathbb{1}$ is nonzero by the triangle identity, and hence surjective because the unit is simple. Exactness and $F(\mathbb{1}) \simeq \mathbb{1}$ imply $F(X) \neq 0$. If f is a nonzero morphism, its image is nonzero. Exactness gives $\operatorname{im} F(f) \simeq F(\operatorname{im} f) \neq 0$, so $F(f) \neq 0$. ∎

8.2 Lemma. *A symmetric tensor functor preserves categorical traces and dimensions.*

Proof. Applying F to evaluation and coevaluation, and inserting F_0, F_2, makes $F(X^*)$ a dual of $F(X)$. The triangle identities follow by applying F to those for X. Symmetry compatibility identifies the image of the braiding in the trace formula with the braiding of $F(X)$ and $F(X^*)$. Thus

$$\operatorname{Tr}(F(f)) = F_0^{-1} F(\operatorname{Tr}(f)) F_0.$$

On the scalar endomorphisms of the unit this is the same scalar because F is k-linear. Set $f = \mathrm{id}_X$ for the dimension statement. ∎

For $\operatorname{char} k \neq 2$, there is no symmetric tensor functor from sVec to $\operatorname{Rep}_k(G)$ for an ordinary affine group scheme G. Indeed, the odd line Π has tensor square $\mathbb{1}$ and categorical dimension -1. Its image would have one-dimensional underlying space, hence dimension 1, contradicting Lemma 8.2. The tensor-square condition is essential here: in positive characteristic an arbitrary vector space can have dimension congruent to -1 in k.

Moderate growth

Write $\ell(X)$ for the composition length of an object in a finite-length abelian category.

8.3 Definition. An object X has *moderate growth* if there is an integer $d_X \geq 1$ such that

$$\ell(X^{\otimes n}) \leq d_X^n \qquad (n \geq 0).$$

A tensor category has moderate growth if every object does.

8.4 Lemma. *For an exact faithful functor $F : \mathcal{C} \to \mathcal{D}$ between finite-length abelian categories, $\ell(X) \leq \ell(F(X))$.*

Proof. Apply F to a composition series of X. Exactness preserves its successive quotients, and faithfulness makes their images nonzero. Each image has length at least one; additivity of length gives the inequality. ∎

Consequently, moderate growth pulls back along tensor functors. In $\operatorname{Rep}(G, z)$ the ordinary total vector-space dimension, rather than superdimension, gives

$$\ell(X^{\otimes n}) \leq (\dim_k X_0 + \dim_k X_1)^n.$$

Categories with finitely many simple isomorphism classes also have moderate growth. To see this, choose representatives $S_1, \dots, S_r$. For fixed X put

$$d_X = \max\{1, \ell(X \otimes S_1), \dots, \ell(X \otimes S_r)\}.$$

Tensor exactness and a composition series of Y give $\ell(X \otimes Y) \leq d_X \ell(Y)$. Starting with $Y = \mathbb{1}$ proves the required bound by induction. In a finite tensor category one can alternatively use the Frobenius–Perron dimension: its positivity on simples gives $\ell(X^{\otimes n}) \leq \operatorname{FPdim}(X)^n$.

8.5 Theorem (Deligne)**.** *Let k be algebraically closed of characteristic zero and let $\mathcal{C}$ be a symmetric tensor category. The following are equivalent:*

1. *$\mathcal{C}$ has moderate growth;*
2. *there is a symmetric tensor functor $\mathcal{C} \to \mathrm{sVec}_k$;*
3. *$\mathcal{C} \simeq \mathrm{Rep}(G, z)$ as a symmetric tensor category for an affine supergroup G and an element z with $z^2 = 1$ and $\mathrm{Ad}(z)$ equal to parity.*

The implication from moderate growth to a super fiber functor is a substantial theorem, not proved here; see [EGNO15, Theorems 9.11.4 and 9.11.9]. The length estimate above proves the converse. The reconstruction behind the remaining equivalence motivates our next lectures, starting with ordinary fiber functors to vector spaces.

9 Lecture 9 (February 3, 2025, est.): Affine groups and tensor automorphisms

Scribe: Guillermo Sanmarco

Our first reconstruction problem is to recover an affine group scheme G from its finite-dimensional representation category together with the forgetful functor. Throughout this lecture k is a field; no finite-type assumption on G is imposed.

Approximation by algebraic groups

9.1 Proposition. *Every finite subset of a commutative Hopf algebra A over k is contained in a Hopf subalgebra finitely generated as a k-algebra.*

Proof. By local finiteness of the regular comodule, the given subset lies in a finite-dimensional subcomodule $V \subseteq A$. Write $\Delta(v_j) = \sum_i v_i \otimes a_{ij}$ in a basis of V. Coassociativity gives

$$\Delta(a_{ij}) = \sum_s a_{is} \otimes a_{sj}.$$

Thus the finite-dimensional span C of the v_j, the a_{ij}, and 1 satisfies $\Delta(C) \subseteq C \otimes C$. Let A_0 be the algebra generated by $C + S(C)$. Since the antipode reverses coproducts, both $\Delta(C)$ and $\Delta(S(C))$ lie in $A_0 \otimes A_0$. For a commutative Hopf algebra $S^2 = \mathrm{id}$, so $S(A_0) \subseteq A_0$. The counit restricts, and A_0 is the desired Hopf subalgebra. ∎

The commutativity assumption matters for this short argument: it is not enough, for a completely arbitrary Hopf algebra, merely to add one image under the antipode.

9.2 Corollary. *Every affine group scheme is an inverse limit of affine algebraic group schemes*

$$G \simeq \varprojlim_i G_i$$

with faithfully flat transition morphisms.

Proof. Write $A = \mathcal{O}(G) = \bigcup_i A_i$, directed by inclusion over its finitely generated Hopf subalgebras. Then $G_i = \operatorname{Spec} A_i$ is algebraic, and

$$\mathrm{Hom}_{k\text{-alg}}(A, R) \simeq \varprojlim_i \mathrm{Hom}_{k\text{-alg}}(A_i, R)$$

for every commutative k-algebra R. An inclusion of commutative Hopf algebras over a field is faithfully flat; applying this to $A_i \subseteq A_j$ gives the final assertion. This faithful-flatness theorem is proved in [Mil12, Chapter VI, Section 11]. ∎

The algebras A_i are finitely generated, not necessarily finite dimensional. Also, faithful flatness of $G_j \to G_i$ does not assert surjectivity of $G_j(R) \to G_i(R)$ for every test algebra R.

The tensor automorphism functor

Let $\omega : \mathrm{Rep}_k(G) \to \mathrm{Vec}_k$ forget the action. For a commutative k-algebra R, put $\omega_R(V) = \omega(V) \otimes_k R$.

9.3 Definition. The group $\underline{\mathrm{Aut}}^{\otimes}(\omega)(R)$ consists of families of invertible R-linear maps $\lambda_V : \omega_R(V) \to \omega_R(V)$ satisfying

1. $(f \otimes \mathrm{id}_R)\lambda_V = \lambda_W(f \otimes \mathrm{id}_R)$ for every morphism $f : V \to W$;
2. $\lambda_{V\otimes W} = \lambda_V \otimes_R \lambda_W$ under the canonical tensor identification;
3. $\lambda_k = \mathrm{id}_R$ on the trivial representation.

Composition is componentwise; scalar extension makes this a group-valued functor of R.

An element $g \in G(R)$ acts on every V_R, and equivariance of morphisms and the diagonal tensor action give a natural group homomorphism

$$G \longrightarrow \underline{\mathrm{Aut}}^{\otimes}(\omega).$$

We shall prove that it is an isomorphism in Lecture 10.

Reduction to a tensor generator

For $X \in \mathrm{Rep}_k(G)$, let $\mathcal{C}_X$ be the full subcategory of subquotients of finite direct sums of tensor products of X and X^*, including the unit. A tensor automorphism of $\omega|_{\mathcal{C}_X}$ is determined by its component on X. Indeed, the component on X^* is determined by evaluation, those on tensor products and sums follow from tensor compatibility and naturality, and those on subquotients follow from injectivity and surjectivity after tensoring with R. Hence

$$\underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X})(R) \hookrightarrow \mathrm{GL}(X \otimes R).$$

Let G_X be the scheme-theoretic image of $G \to \mathrm{GL}(X)$. It is a closed algebraic subgroup. Its coordinate Hopf algebra is generated by the matrix coefficients of X and its dual. A faithful finite-dimensional representation generates the representation category of an algebraic group under tensor products, sums, duals and subquotients; see [Mil12, Chapter VIII, Section 11]. Thus $\mathcal{C}_X$ identifies with $\mathrm{Rep}(G_X)$, and its action gives inclusions

$$G_X(R) \subseteq \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X})(R) \subseteq \mathrm{GL}(X \otimes R).$$

Naturality gives preservation of every subrepresentation and compatibility with every quotient. For example, if $Y \subseteq V$, the inclusion gives

$$\lambda_V(Y \otimes R) = Y \otimes R.$$

If $t \in V^G$, the morphism $k \to V$, $1 \mapsto t$, gives $\lambda_V(t \otimes 1) = t \otimes 1$.

Finally, as X varies and direct sums provide upper bounds, the Hopf algebras $\mathcal{O}(G_X)$ exhaust $\mathcal{O}(G)$. Indeed, every element belongs to a finite-dimensional regular

subcomodule and is a linear combination of its matrix coefficients by the counit identity. Thus

$$G = \varprojlim_X G_X, \qquad \underline{\mathrm{Aut}}^{\otimes}(\omega) = \varprojlim_X \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X}).$$

The second equality holds because every finite collection of representations and morphisms lies in some $\mathcal{C}_X$.

10 Lecture 10 (February 5, 2025): Reconstruction of an affine group

Scribe: Jackson Morris

We prove the reconstruction statement announced in Lecture 9.

10.1 Theorem. *For an affine group scheme G over a field k, the natural morphism*

$$G \longrightarrow \underline{\mathrm{Aut}}^{\otimes}(\omega), \qquad \omega : \mathrm{Rep}_k(G) \to \mathrm{Vec}_k,$$

is an isomorphism of group-valued functors. In particular, the tensor automorphism functor is representable.

Chevalley's theorem

10.2 Theorem (Chevalley)**.** *Every closed subgroup scheme H of an affine algebraic group K is the stabilizer of a one-dimensional subspace in a finite-dimensional representation of K.*

See [Mil12, Chapter VIII, Theorem 13.1].

Proof of Theorem 10.1. Fix $X \in \mathrm{Rep}_k(G)$ and put $H = G_X$ and $K = \mathrm{GL}(X)$. As in Lecture 9, the tensor subcategory $\mathcal{C}_X$ generated by X identifies with $\mathrm{Rep}(H)$, and for every commutative k-algebra R we have inclusions

$$H(R) \subseteq \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X})(R) \subseteq K(R).$$

Let λ be a tensor automorphism in the middle group and write $g = \lambda_X \in K(R)$. By Chevalley's theorem, choose a K-representation W and a line $L \subseteq W$ with $H = \mathrm{Stab}_K(L)$.

Every finite-dimensional K-representation is a subquotient of a finite direct sum of tensor products of X and X^* [Mil12, Chapter VIII, Theorem 11.7]. Thus $W|_H$ belongs to $\mathcal{C}_X$, and the action of g on $W \otimes R$ agrees with λ_W: both are obtained from g by duals, tensor products, direct sums and passage to subquotients.

The line L is an H-subrepresentation of W. Naturality for its inclusion $i : L \hookrightarrow W$ gives

$$\begin{array}{ccc} L \otimes R & \xrightarrow{\lambda_L} & L \otimes R \\ {\scriptstyle i\otimes 1}\downarrow & & \downarrow{\scriptstyle i\otimes 1} \\ W \otimes R & \xrightarrow[\lambda_W]{} & W \otimes R. \end{array}$$

Since λ_L is invertible, $g(L \otimes R) = L \otimes R$. Hence $g \in H(R)$. Its action on X is λ_X, which determines all components on $\mathcal{C}_X$, so its action is exactly λ. This proves

$$G_X \simeq \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X}).$$

These isomorphisms commute with restriction when $\mathcal{C}_X \subseteq \mathcal{C}_{X\oplus Y}$. The Hopf algebras $\mathcal{O}(G_X)$ are finitely generated as k-algebras and exhaust $\mathcal{O}(G)$, as proved

in Lecture 9. Passing to the compatible inverse limits gives

$$G \simeq \varprojlim_X G_X \simeq \varprojlim_X \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X}) \simeq \underline{\mathrm{Aut}}^{\otimes}(\omega).$$

All constructions are natural in R and respect the group operations. ∎

Functoriality

A group homomorphism $f : G \to G'$ gives restriction of representations

$$f^* : \mathrm{Rep}(G') \longrightarrow \mathrm{Rep}(G),$$

with the canonical identification $\omega_G f^* = \omega_{G'}$.

10.3 Corollary. *Group homomorphisms $G \to G'$ correspond to symmetric tensor functors $F : \mathrm{Rep}(G') \to \mathrm{Rep}(G)$ equipped with a specified tensor isomorphism $\alpha : \omega_G F \simeq \omega_{G'}$, up to isomorphisms respecting α.*

Proof. For each R, an element $g \in G(R)$ acts on every $F(V)_R$. Transport this action through α to obtain a tensor automorphism of $\omega_{G',R}$. Theorem 10.1 identifies it with a unique element of $G'(R)$. Naturality and multiplication give a group-scheme map $f : G \to G'$, and the same construction shows that α intertwines F with f^*. ∎

When k is algebraically closed, every symmetric tensor functor $F : \mathrm{Rep}(G') \to \mathrm{Rep}(G)$ is isomorphic to restriction along a group homomorphism $G \to G'$. Indeed, the two fiber functors $\omega_G F$ and $\omega_{G'}$ are tensor-isomorphic [Wib25, Corollary 1.2]; choosing such an isomorphism and applying the preceding corollary gives the homomorphism.

The general reconstruction problem

We next prove the following extension, in Lectures 11–13.

10.4 Theorem (Neutral Tannaka reconstruction)**.** *Let $\mathcal{C}$ be an essentially small symmetric tensor category over k, and let $\omega : \mathcal{C} \to \mathrm{Vec}_k$ be an exact faithful k-linear strong symmetric monoidal functor to finite-dimensional vector spaces. Then*

$$G = \underline{\mathrm{Aut}}^{\otimes}(\omega)$$

is an affine group scheme, and ω lifts to a symmetric tensor equivalence $\widetilde{\omega} : \mathcal{C} \simeq \mathrm{Rep}_k(G)$ whose composition with the forgetful functor is ω.

We first reconstruct a coalgebra from the abelian category and ω, then use the tensor product, symmetry and duals to recover multiplication, commutativity and the antipode.

11 Lecture 11 (February 7, 2025, est.): Abelian reconstruction and transporters

Scribe: Guillermo Sanmarco

For the first part of reconstruction, forget the tensor structure. Let $\mathcal{C}$ be an essentially small k-linear abelian category and

$$\omega : \mathcal{C} \longrightarrow \mathrm{Vec}_k$$

an exact faithful k-linear functor, with finite-dimensional values. We will construct a coalgebra B and an equivalence $\mathcal{C} \simeq \mathrm{Comod}_{\mathrm{fd}}(B)$ compatible with ω.

Finite-dimensional algebras and coalgebras

If A is a finite-dimensional algebra, its dual $A^* = \mathrm{Hom}_k(A, k)$ is a coalgebra, with

$$\Delta_{A^*}(\phi)(a \otimes b) = \phi(ab), \qquad \varepsilon_{A^*}(\phi) = \phi(1).$$

A finite-dimensional left A-module M becomes a right A^*-comodule. For a basis a_i of A with dual basis a_i^*, its coaction is

$$m \longmapsto \sum_i a_i m \otimes a_i^*.$$

This is independent of the basis. Associativity of the action is coassociativity of the coaction, and the unit condition is the counit identity. Conversely, a coaction gives $a \cdot m = (\mathrm{id} \otimes \mathrm{ev}_a)\rho_M(m)$. These constructions are inverse and preserve underlying vector spaces.

Tensoring an object by a vector space

A k-linear abelian category admits an action of finite-dimensional vector spaces. For $V = k^n$, define $V \otimes X = X^{\oplus n}$ and let matrices act by the corresponding matrices of morphisms. Choosing bases extends the construction to all finite-dimensional V; changes of basis give canonical comparison isomorphisms. Equivalently, the object is characterized by

$$\mathrm{Hom}_{\mathcal{C}}(V \otimes X, Y) \simeq \mathrm{Hom}_k\big(V, \mathrm{Hom}_{\mathcal{C}}(X, Y)\big).$$

Here the tensor product is this vector-space action, not a monoidal structure on $\mathcal{C}$. There is also a natural isomorphism

$$\mathrm{Hom}_{\mathcal{C}}(Y, V \otimes X) \simeq V \otimes_k \mathrm{Hom}_{\mathcal{C}}(Y, X).$$

Write

$$\underline{\mathrm{Hom}}(V, X) = V^* \otimes X.$$

A k-linear functor commutes with these constructions up to their natural identifications; in particular

$$\omega\big(\underline{\mathrm{Hom}}(V, X)\big) \simeq \mathrm{Hom}_k(V, \omega X).$$

11.1 Definition. For a subspace $W \subseteq V$ and a subobject $Y \subseteq X$, the *transporter* of W into Y is

$$(Y : W) = \ker\big(\underline{\mathrm{Hom}}(V, X) \longrightarrow \underline{\mathrm{Hom}}(W, X/Y)\big).$$

Exactness gives

$$\omega(Y : W) = \{f : V \to \omega X \mid f(W) \subseteq \omega Y\}.$$

Thus linear conditions about preserving subspaces can be expressed by subobjects in $\mathcal{C}$.

The object of endomorphisms preserving subobjects

Faithfulness implies $\omega X = 0 \Rightarrow X = 0$. If $Y \subsetneq X$, exactness gives $\dim \omega Y < \dim \omega X$. Hence every object of $\mathcal{C}$ has finite length, and every descending chain of subobjects terminates. In particular, intersections of arbitrarily many subobjects of a fixed object are already finite intersections. Exactness of ω then preserves these intersections. It also reflects inclusions between subobjects of a common object: if $\omega P \subseteq \omega Q$, apply ω to $P \cap Q \subseteq P$ to conclude $P \subseteq Q$.

Fix X, put $V = \omega X$ and $H_X = \underline{\mathrm{Hom}}(V, X)$. For each $n \geq 1$, let

$$d_n : H_X \longrightarrow \underline{\mathrm{Hom}}(V^{\oplus n}, X^{\oplus n})$$

be the diagonal map. Define

$$P_X = \bigcap_{n \geq 1,\, Y \subseteq X^{\oplus n}} d_n^{-1}(Y : \omega Y) \subseteq H_X.$$

The intersection exists by the preceding finite-length argument. Its image under ω is

$$A_X := \omega(P_X) = \{a \in \mathrm{End}_k(V) \mid a^{\oplus n}(\omega Y) \subseteq \omega Y \text{ for all } Y \subseteq X^{\oplus n},\ n \geq 1\}.$$

This is a finite-dimensional unital subalgebra: identities preserve all the subspaces, and preservation is closed under linear combinations and composition.

11.2 Lemma. *The object P_X is the smallest subobject $Q \subseteq H_X$ such that* $\mathrm{id}_V \in \omega Q \subseteq \mathrm{End}_k(V)$.

Proof. The identity belongs to A_X, so P_X is one such subobject. Conversely, suppose $Q \subseteq H_X$ and $\mathrm{id}_V \in \omega Q$. Choose a basis of V^* to identify H_X with $X^{\oplus d}$, $d = \dim V$. By the defining property of A_X, the diagonal action of every $a \in A_X$ preserves ωQ. Under $\omega H_X = \mathrm{End}_k(V)$, this diagonal action is left composition, $q \mapsto a \circ q$. Applying it to id_V shows $a \in \omega Q$. Thus $\omega P_X \subseteq \omega Q$, and reflection of subobject inclusions gives $P_X \subseteq Q$. ∎

Let $\langle X \rangle_{\oplus}$ denote the full subcategory consisting of subquotients of finite direct sums of X. This notation is different from the tensor-generated category $\mathcal{C}_X$ used

earlier. The category $\langle X\rangle_\oplus$ is abelian, and its inclusion in $\mathcal{C}$ is exact: kernels, cokernels and finite sums of such subquotients are again such subquotients.

Each $a \in A_X$ acts diagonally on $\omega(X^{\oplus n})$, preserves every subobject, and hence acts on the image under ω of every subquotient. These actions are independent of presentations and natural in $Y \in \langle X\rangle_\oplus$. To check this, take the graph of a morphism between two subquotients, pull it back to their presenting subobjects, and regard that pullback as a subobject of a direct sum of copies of X. Its preservation is exactly the assertion that the morphism intertwines the two actions. In particular, apply this to an identity morphism to compare presentations.

We have therefore lifted ω to an exact faithful functor

$$\widetilde{\omega}_X : \langle X\rangle_\oplus \longrightarrow A_X\text{-mod}_{\mathrm{fd}}.$$

The next lecture proves that it is an equivalence and that $A_X = \mathrm{End}(\omega|_{\langle X\rangle_\oplus})$. The construction here follows the abelian part of [Mil12, Chapter X, Section 3], with the transporter and minimal-subobject arguments made explicit.

12 Lecture 12 (February 10, 2025, est.): From a fiber functor to a coalgebra

Scribe: Guillermo Sanmarco

Continue with the exact faithful k-linear functor $\omega : \mathcal{C} \to \mathrm{Vec}_k$ and the objects P_X and algebras A_X of Lecture 11.

The local equivalence

12.1 Theorem. *The functor $\widetilde{\omega}_X$ is an equivalence*

$$\langle X\rangle_{\oplus} \simeq A_X\text{-mod}_{\mathrm{fd}}, \qquad A_X \simeq \mathrm{End}(\omega|_{\langle X\rangle_{\oplus}}).$$

Proof. Write $A = A_X$, $P = P_X$, and $V = \omega X$. Precomposition by $a \in A$ defines a morphism on $H_X = V^* \otimes X$ whose image under ω sends b to ba. It preserves P, because $\omega P = A$ is closed under multiplication and ω reflects subobject inclusions. These maps make P a right A-module object in $\mathcal{C}$. Its image under ω is the right regular module. Its natural left A-action under $\widetilde{\omega}_X$ is left multiplication, so the two actions commute.

For a finite-dimensional left A-module M, form the object

$$Q(M) = P \otimes_A M = \mathrm{coker}\big(P \otimes A \otimes M \xrightarrow{r\otimes \mathrm{id} - \mathrm{id}\otimes \mu_M} P \otimes M\big).$$

The products with vector spaces are the external products of Lecture 11. This cokernel belongs to $\langle X\rangle_{\oplus}$, and exactness gives, naturally and as left A-modules,

$$\widetilde{\omega}_X Q(M) \simeq A \otimes_A M \simeq M.$$

To identify the other composite, start with the evaluation morphism

$$P \otimes \omega X \longrightarrow X,$$

obtained by restricting evaluation from $H_X \otimes V$. For $X^{\oplus n}$ take its direct sum. If $Y \subseteq X^{\oplus n}$, its restriction to $P \otimes \omega Y$ factors through Y: after applying ω, this says precisely that A preserves ωY. Exact faithfulness detects the required zero composite to the quotient. Passing to quotients defines evaluation for every subquotient $Y \in \langle X\rangle_{\oplus}$. The graph argument from the previous lecture makes these evaluations independent of the presentation and natural in Y.

They are balanced over A, so they induce

$$Q(\widetilde{\omega}_X Y) = P \otimes_A \omega Y \longrightarrow Y.$$

Applying ω gives the isomorphism $A \otimes_A \omega Y \to \omega Y$. An exact faithful functor reflects isomorphisms, by applying it to kernel and cokernel. Thus this evaluation is an isomorphism. The two natural isomorphisms exhibit Q as a quasi-inverse, proving the equivalence.

Finally, A acts naturally on all these vector spaces, giving a map to the endomorphism algebra of the restricted functor. Conversely, a natural endomorphism is determined by its component on X, since all objects are subquotients of finite sums of X. Naturality forces that component to preserve every $\omega Y \subseteq \omega(X^{\oplus n})$, so it lies in A. These constructions are inverse algebra maps. ∎

Passing from local algebras to one coalgebra

Set $B_X = A_X^*$. The preceding theorem and algebra–coalgebra duality give

$$\langle X\rangle_\oplus \simeq \mathrm{Comod}_{\mathrm{fd}}(B_X).$$

If $\langle X\rangle_\oplus \subseteq \langle Z\rangle_\oplus$, restriction of natural endomorphisms gives $A_Z \to A_X$. This map is surjective. Indeed, under the local equivalence for Z, put $M = \omega X$ and $D = A_Z/\operatorname{Ann}(M)$. The D-module M is faithful. Choosing a basis $m_1, \dots, m_d$ embeds the regular module by

$$D \longrightarrow M^{\oplus d}, \qquad a \longmapsto (am_1, \dots, am_d).$$

Every finite-dimensional D-module is a quotient of a finite direct sum of copies of D, hence a subquotient of copies of M. Thus the subcategory generated by M is exactly D-$\mathrm{mod}_{\mathrm{fd}}$, and $A_X \simeq D$. Dualizing gives an injection $B_X \hookrightarrow B_Z$.

Direct sums of objects make these subcategories a directed system. Define

$$B = \varinjlim_X B_X = \bigcup_X B_X$$

with the compatible coalgebra structures. Every object and every finite collection of morphisms of $\mathcal{C}$ belongs to some $\langle X\rangle_\oplus$. The local equivalences therefore assemble into a fully faithful functor $\mathcal{C} \to \mathrm{Comod}_{\mathrm{fd}}(B)$. It is essentially surjective: the finitely many matrix coefficients of a finite-dimensional B-comodule lie in some B_X, so its coaction already factors through that subcoalgebra. We have proved the abelian reconstruction theorem

$$\mathcal{C} \simeq \mathrm{Comod}_{\mathrm{fd}}(B), \tag{12.1}$$

compatibly with ω.

Natural endomorphisms and matrix coefficients

Let A now be any finite-dimensional algebra and let $U : A\text{-}\mathrm{mod}_{\mathrm{fd}} \to \mathrm{Vec}_k$ forget the action. For a commutative algebra R, put $U_R = U(-) \otimes R$. There is an algebra isomorphism

$$A \otimes R \simeq \mathrm{End}(U_R).$$

One direction lets $A \otimes R$ act. For the other, send a natural endomorphism λ to $\lambda_A(1 \otimes 1)$. For $m \in M$, the A-linear map $A \to M$, $a \mapsto am$, gives

$$\lambda_M(m \otimes 1) = \lambda_A(1 \otimes 1) \cdot m.$$

This proves that the two constructions are inverse. Notice that the components of λ need only be R-linear, not A-linear.

Applying this locally and passing to inverse limits yields

$$\mathrm{End}(\omega_R) \simeq \varprojlim_X (A_X \otimes R) \simeq \mathrm{Hom}_k(B, R).$$

The multiplication on the right is convolution using the coproduct of B.

It is useful to describe B without the directed system. For $v \in \omega X$ and $\phi \in (\omega X)^*$, write $[\phi \otimes v]_X$ for a matrix coefficient. Then

$$B \simeq \left(\bigoplus_X (\omega X)^* \otimes \omega X \right) / \mathcal{R},$$

where a small skeleton is used and $\mathcal{R}$ is spanned by

$$[\phi \otimes \omega(f)v]_Y - [\phi \circ \omega(f) \otimes v]_X, \qquad f : X \to Y.$$

Indeed, a linear map from this quotient to any vector space encodes precisely a natural family of linear maps on the spaces ωX with values in their tensor products with that vector space. On each local subcategory the preceding calculation identifies the quotient with A_X^*; taking the directed union gives B. If e_i is a basis of ωX with dual basis e_i^*, the coalgebra structure and universal coaction are

$$\begin{aligned} \Delta[\phi \otimes v]_X &= \sum_i [\phi \otimes e_i]_X \otimes [e_i^* \otimes v]_X, \\ \varepsilon[\phi \otimes v]_X &= \phi(v), \\ \rho_X(v) &= \sum_i e_i \otimes [e_i^* \otimes v]_X. \end{aligned}$$

The expressions are independent of the basis by the canonical tensor representing the identity.

In particular, applying this construction to the forgetful functor on finite-dimensional comodules of a coalgebra C recovers C: its finite-dimensional coefficient subcoalgebras exhaust it. A coalgebra map $C \to C'$ gives a functor on comodules by composing coactions. Conversely, a functor compatible with the forgetful functors sends the coefficient $[\phi \otimes v]_X$ to $[\phi \otimes v]_{F(X)}$; naturality makes this well-defined and the formulas above show it is a coalgebra map. This is the functorial form of abelian reconstruction used next.

13 Lecture 13 (February 12, 2025, est.): Completing Tannaka reconstruction

Scribe: Guillermo Sanmarco

We now restore the symmetric tensor structure on $\mathcal{C}$. Let $\omega : \mathcal{C} \to \mathrm{Vec}_k$ be a symmetric tensor functor. The preceding lectures construct a coalgebra B and an equivalence

$$\mathcal{C} \simeq \mathrm{Comod}_{\mathrm{fd}}(B)$$

compatible with the underlying vector spaces. We show that the tensor product defines multiplication on B, symmetry makes it commutative, and rigidity supplies the antipode.

Tensor products of comodules

Suppose first that B is merely a coalgebra. A coalgebra homomorphism $m : B \otimes B \to B$ equips the vector space $V \otimes W$ with the coaction

$$v \otimes w \longmapsto \sum v_{(0)} \otimes w_{(0)} \otimes m(v_{(1)} \otimes w_{(1)}).$$

Coassociativity and the counit identity follow from the fact that m is a coalgebra map. This construction is functorial and bilinear on morphisms. Conversely, a bilinear bifunctor on finite-dimensional B-comodules whose underlying vector-space functor is the ordinary tensor product determines a unique such m. Explicitly, the tensor product of two matrix coefficients is sent to the corresponding coefficient of the specified coaction on $V \otimes W$. The naturality relations of Lecture 12, separately in the two variables, make this well-defined, and the coefficient coproduct formula makes it a coalgebra homomorphism.

The usual vector-space associator is a comodule map exactly when m is associative: on a triple tensor product the two coactions have coefficients $m(m(b \otimes c) \otimes d)$ and $m(b \otimes m(c \otimes d))$. Since coefficients span B, equality on all triple tensor products is equality of those maps. Similarly, the ordinary flip is a comodule map exactly when m is commutative. A unit object with underlying space k determines a coalgebra map $u : k \to B$; the usual unit constraints are comodule maps precisely when u is a two-sided unit for m.

A lemma about rigidity

13.1 Lemma. *Let $\mathcal{C}$ be rigid and let $F, G : \mathcal{C} \to \mathcal{D}$ be strong monoidal functors. Every monoidal natural transformation $\alpha : F \to G$ is invertible.*

Proof. Strong monoidal functors preserve duals. Use the induced identifications $F(X^*) \simeq F(X)^*$ and $G(X^*) \simeq G(X)^*$, and suppress associators and unit constraints. Define

$$\begin{aligned}\beta_X : G(X) &\xrightarrow{\mathrm{coev}_{F(X)} \otimes \mathrm{id}} F(X) \otimes F(X)^* \otimes G(X) \\ &\xrightarrow{\mathrm{id} \otimes \alpha_{X^*} \otimes \mathrm{id}} F(X) \otimes G(X)^* \otimes G(X) \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_{G(X)}} F(X).\end{aligned}$$

Naturality with evaluation, together with the tensor and unit identities for α, gives

$$\mathrm{ev}_{G(X)}(\alpha_{X^*} \otimes \alpha_X) = \mathrm{ev}_{F(X)}\,.$$

It follows from the first triangle identity that $\beta_X \alpha_X = \mathrm{id}_{F(X)}$. Naturality with coevaluation gives

$$(\alpha_X \otimes \alpha_{X^*})\,\mathrm{coev}_{F(X)} = \mathrm{coev}_{G(X)},$$

so the other composite is $\alpha_X \beta_X = \mathrm{id}_{G(X)}$. Thus all components are invertible, and their inverses automatically form a natural monoidal transformation. ∎

No rigidity assumption on every object of $\mathcal{D}$ is needed: the objects in the images of the two functors already have the required duals.

The reconstructed Hopf algebra

We prove Theorem 10.4. Write matrix coefficients in the notation of Lecture 12, and use the monoidal identifications of ω to suppress its tensor constraints. Define

$$[\phi \otimes v]_X[\psi \otimes w]_Y = [(\phi \otimes \psi) \otimes (v \otimes w)]_{X\otimes Y}, \qquad 1_B = [1^* \otimes 1]_{\mathbb{1}},$$

where $1^* : k \to k$ is the identity. Naturality of the tensor product respects the coefficient relations, so multiplication is well-defined. The associator and unit constraints give associativity and the unit identities. The symmetric braiding maps under ω to the ordinary flip, so its naturality relation identifies the two orders of the product. Thus B is a commutative algebra.

Using tensor products of bases in the coefficient formula gives

$$\Delta(bc) = \Delta(b)\Delta(c), \qquad \varepsilon(bc) = \varepsilon(b)\varepsilon(c),$$

and $\Delta(1_B) = 1_B \otimes 1_B$, $\varepsilon(1_B) = 1$. Therefore B is a commutative bialgebra. By construction, its comodule tensor product agrees with the one transported from $\mathcal{C}$.

For each commutative k-algebra R, Lecture 12 identifies a linear map $g : B \to R$ with the natural endomorphism

$$\lambda_X(v \otimes 1) = \sum_i e_i \otimes g([e_i^* \otimes v]_X)$$

of ω_R. The multiplication and unit formulas show that

$$\lambda_{X\otimes Y} = \lambda_X \otimes_R \lambda_Y, \quad \lambda_{\mathbb{1}} = \mathrm{id}_R \quad \Longleftrightarrow \quad g(bc) = g(b)g(c), \quad g(1_B) = 1_R.$$

Consequently,

$$\underline{\mathrm{End}}^{\otimes}(\omega)(R) \simeq \mathrm{Hom}_{k\text{-alg}}(B, R).$$

Rigidity and Lemma 13.1, applied to ω_R, show that every such tensor endomorphism is invertible. The functor represented by B is therefore a group functor, and its inverse is represented by an antipode $S : B \to B$.

One can write the antipode explicitly. Identify $\omega(X^*)$ with $(\omega X)^*$, and for $v \in \omega X$ let $\widehat{v} \in (\omega X)^{**}$ be evaluation at v. Then

$$S([\phi \otimes v]_X) = [\widehat{v} \otimes \phi]_{X^*}.$$

The evaluation and coevaluation identities give

$$m(S \otimes \mathrm{id})\Delta = u\varepsilon = m(\mathrm{id} \otimes S)\Delta,$$

which are also the identities expressing that the action on the dual is the inverse transpose. This realizes the group-functor inverse just constructed.

Set $G = \operatorname{Spec} B$. We have proved, naturally in R,

$$G(R) = \underline{\mathrm{Aut}}^{\otimes}(\omega)(R).$$

Equation (12.1) now becomes

$$\widetilde{\omega} : \mathcal{C} \simeq \mathrm{Comod}_{\mathrm{fd}}(\mathcal{O}(G)) = \mathrm{Rep}_k(G).$$

The construction of multiplication and symmetry makes this an equivalence of symmetric tensor categories, and its composite with the forgetful functor is ω. This proves neutral Tannaka reconstruction.

13.2 Remark. The pair $(\mathcal{C}, \omega)$ determines G. Keeping the fiber functor in the statement is essential: the category without a chosen fiber functor does not specify this identification. For comparison with the classical formulation, see [Mil12, Chapter X, Section 3] and [EGNO15, Section 5.4].

14 Lecture 14 (February 14, 2025, est.): Relative Tannaka duality

Scribe: Guillermo Sanmarco

14.1 Changing the target of a fiber functor

The next goal is to study symmetric tensor categories by realizing them as categories of representations of affine group schemes in a suitable symmetric tensor category. Ordinary Tannaka duality uses Vect_k as the target; the relative version allows a different target $\mathcal{D}$.

Fix a perfect field k. In this lecture, a symmetric tensor category is essentially small, k-linear, abelian and rigid, with finite-dimensional morphism spaces, objects of finite length, and $\mathrm{End}(\mathbb{1}) = k$. A tensor functor is exact, k-linear and strong symmetric monoidal. Such a functor is faithful. Write Symm_k for the category of these categories and isomorphism classes of tensor functors. Its full subcategory of categories of moderate growth consists of those $\mathcal{C}$ for which, for each $X \in \mathcal{C}$, there is a real number $a_X \geq 1$ satisfying

$$\mathrm{length}(X^{\otimes n}) \leq a_X^n \qquad (n \geq 1).$$

Fix $\mathcal{D} \in \mathrm{Symm}_k$. An object of $\mathrm{Symm}(\mathcal{D})$ is a pair $(\mathcal{C}, F)$ with $F : \mathcal{C} \to \mathcal{D}$ a tensor functor. A morphism from $(\mathcal{C}_1, F_1)$ to $(\mathcal{C}_2, F_2)$ is represented by a pair (T, α), where $T : \mathcal{C}_1 \to \mathcal{C}_2$ is a tensor functor and $\alpha : F_1 \xrightarrow{\sim} F_2 T$ is a monoidal natural isomorphism. We identify two such pairs when there is a compatible monoidal natural isomorphism between their functors T. Because F_2 is faithful, a compatible isomorphism between two fixed pairs is unique if it exists. This does not say that the morphisms of $\mathrm{Symm}(\mathcal{D})$ themselves are unique.

14.2 The internal group of tensor automorphisms

An affine group scheme in $\mathcal{D}$ is described by a commutative Hopf algebra in $\mathrm{Ind}(\mathcal{D})$. The ind-completion is necessary: a coordinate algebra need not be an object of finite length. If $F : \mathcal{C} \to \mathcal{D}$ is a tensor functor, the coefficient construction gives a commutative Hopf algebra

$$A_F = \int^{X \in \mathcal{C}} F(X)^* \otimes F(X) \quad \text{in } \mathrm{Ind}(\mathcal{D}).$$

Here the integral denotes a coend. Concretely, its structure maps $i_X : F(X)^* \otimes F(X) \to A_F$ satisfy, for every $f : X \to Y$,

$$i_X\big(F(f)^* \otimes \mathrm{id}_{F(X)}\big) = i_Y\big(\mathrm{id}_{F(Y)^*} \otimes F(f)\big) \quad \text{on } F(Y)^* \otimes F(X).$$

It is universal for families with these relations. Tensor products in $\mathcal{C}$ define multiplication, the unit object defines the unit, evaluation and coevaluation define the coalgebra structure, and duality defines the antipode. Symmetry makes the multiplication commutative. This is the internal analogue of the matrix-coefficient Hopf algebra.

14.1 Definition. The affine group scheme represented by A_F is denoted $\underline{\mathrm{Aut}}^{\otimes}(F)$. The *fundamental group* of $\mathcal{D}$ is

$$\pi(\mathcal{D}) = \underline{\mathrm{Aut}}^{\otimes}(\mathrm{Id}_{\mathcal{D}}).$$

The universal coefficient coactions make every object of $\mathcal{D}$, and then every ind-object, a representation of $\pi(\mathcal{D})$. Naturality means that every morphism in $\mathcal{D}$ respects these canonical actions. Applying the canonical action to the objects $F(X)$ gives a homomorphism

$$\psi_F : \pi(\mathcal{D}) \longrightarrow \underline{\mathrm{Aut}}^{\otimes}(F).$$

14.2 Example. We have $\pi(\mathrm{Vect}_k) = 1$. Indeed, naturality with respect to maps $k \to V$, together with the tensor-unit condition, forces a tensor automorphism of the identity to be the identity on every vector space. If $\operatorname{char} k \neq 2$, then $\pi(\mathrm{SVec}_k) = \mu_2 \cong \mathbb{Z}/2$, regarded as an internal group concentrated in even degree. Its nontrivial element acts as $+1$ on the even part and -1 on the odd part. This is the parity automorphism, not the functor that exchanges even and odd degrees.

14.3 Compatible groups and representations

Not every homomorphism from $\pi(\mathcal{D})$ to an internal group is compatible with the structure already present in $\mathcal{D}$.

14.3 Definition. A *$\mathcal{D}$-group* is a pair (G, ϕ), where G is an affine group scheme in $\mathcal{D}$ and $\phi : \pi(\mathcal{D}) \to G$ is a homomorphism such that the canonical action of $\pi(\mathcal{D})$ on $\mathcal{O}(G)$ equals the action induced by conjugation in G along ϕ. A morphism $(G, \phi) \to (H, \psi)$ is a homomorphism $u : G \to H$ with $u\phi = \psi$.

For clarity, put $A = \mathcal{O}(G)$ and $B = \mathcal{O}(\pi(\mathcal{D}))$. Using left coactions, the compatibility condition says that the canonical map $A \to B \otimes A$ is the composite

$$A \xrightarrow{\Delta^{(2)}} A \otimes A \otimes A \xrightarrow{\phi^* \otimes \mathrm{id} \otimes S_B \phi^*} B \otimes A \otimes B \xrightarrow{(m_B \otimes \mathrm{id})(\mathrm{id} \otimes c_{A,B})} B \otimes A.$$

The last arrow uses the symmetry to bring the two B factors together. The pair $(\underline{\mathrm{Aut}}^{\otimes}(F), \psi_F)$ is a $\mathcal{D}$-group.

14.4 Definition. For a $\mathcal{D}$-group (G, ϕ), let $\mathrm{Rep}_{\mathcal{D}}(G, \phi)$ be the full subcategory of representations of G on objects of $\mathcal{D}$ whose restriction along ϕ is the canonical $\pi(\mathcal{D})$-action. It has a forgetful tensor functor $\omega : \mathrm{Rep}_{\mathcal{D}}(G, \phi) \to \mathcal{D}$.

14.5 Theorem (Relative Tannaka duality)**.** *The assignments*

$$(\mathcal{C}, F) \longmapsto (\underline{\mathrm{Aut}}^{\otimes}(F), \psi_F), \qquad (G, \phi) \longmapsto (\mathrm{Rep}_{\mathcal{D}}(G, \phi), \omega)$$

give mutually inverse contravariant *equivalences between* $\mathrm{Symm}(\mathcal{D})$ *and the category of $\mathcal{D}$-groups.*

This is the relative reconstruction theorem; see [CEO24, §4, Theorem 4.2.1]. Its full reconstruction proof is not part of this lecture. The variance can be checked directly: a pair (T, α) as above sends an automorphism ξ of F_2 to

$$\alpha^{-1}(\xi T)\alpha \in \underline{\mathrm{Aut}}^{\otimes}(F_1).$$

Thus a functor between representation categories produces a group homomorphism in the opposite direction. For $\mathcal{D} = \mathrm{Vect}_k$, the fundamental group is trivial and we recover ordinary neutral Tannaka duality.

14.6 Example. For $\mathcal{D} = \mathrm{SVec}_k$, $\operatorname{char} k \neq 2$, an internal group is an affine supergroup. A compatible map $\mu_2 \to G$ is specified by an even element $z \in G(k)$ with $z^2 = 1$ whose conjugation action is the parity automorphism of $\mathcal{O}(G)$. The category $\mathrm{Rep}(G, z)$ consists of representations on which z acts by parity. If G is purely even, the condition makes z central. For a general supergroup, centrality must not replace the parity-conjugation condition.

14.4 The characteristic-zero classification

Now suppose that k is algebraically closed of characteristic zero. Deligne's theorem states that a symmetric tensor category admits a fiber functor to SVec_k if and only if it has moderate growth. Consequently it is of the form $\mathrm{Rep}(G, z)$ above. The fiber functor, when it exists, is unique up to monoidal natural isomorphism; the isomorphism need not be unique. These statements apply to arbitrary essentially small pretannakian categories; no tensor-generation hypothesis is imposed. For this formulation and uniqueness, see [CEO24, §§1.1, 1.5]. In the ordinary Tannakian case, uniqueness without a finite-generation hypothesis also follows from [Wib25, Corollary 1.2].

The ordinary Tannakian criterion is stronger: there is a fiber functor to Vect_k if and only if, for every object X, some exterior power $\bigwedge^{n_X} X$ vanishes. The bound may depend on X. The next lecture explains why arbitrary Schur functors replace exterior powers in the super version of this criterion.

15 Lecture 15 (February 19, 2025, est.): Schur functors and Deligne's theorem

Scribe: Guillermo Sanmarco

Throughout this lecture, k is algebraically closed of characteristic zero. We follow the framework of [Del02]. For part of the discussion, one may drop the finite-length and finite-dimensional-Hom conditions: retain an essentially small k-linear abelian rigid symmetric monoidal category $\mathcal{C}$ with $\mathrm{End}(\mathbb{1}) = k$. Tensor product is exact in each variable, because rigidity supplies both adjoints. These are still substantive hypotheses; retaining rigidity alone would not suffice.

15.1 The statement and the finiteness condition

15.1 Definition. An object $X \in \mathcal{C}$ is of *moderate growth* if its tensor powers have finite length and there is $a_X \geq 1$ with

$$\mathrm{length}(X^{\otimes n}) \leq a_X^n \qquad (n \geq 1).$$

It is *Schur-finite* if $S^\lambda(X) = 0$ for some partition λ, where the Schur functor is defined below.

15.2 Theorem (Deligne)**.** *An object is Schur-finite if and only if it is of moderate growth. Schur-finite objects are closed under finite direct sums, tensor products, duals, extensions and subquotients. A symmetric tensor category has a fiber functor to* SVec_k *if and only if all its objects are Schur-finite.*

The existence of the super fiber functor is the deep part of this theorem. This lecture establishes the Schur decomposition and the implication from moderate growth to Schur-finiteness, and records the representation-theoretic tools used in the full proof. The remaining arguments are in [Del02, Proposition 0.5 and Theorem 0.6]; they are not consequences of the growth estimate alone.

15.3 Lemma. *If every object of* $\mathcal{C}$ *has finite length, then every morphism space is finite-dimensional.*

Proof. We use the standard simple-unit lemma for rigid abelian tensor categories: every subobject of $\mathbb{1}$ is a direct summand, so $\mathrm{End}(\mathbb{1}) = k$ implies that $\mathbb{1}$ is simple [DM82, Proposition 1.17]. This is the input used in [Del02, Proposition 1.1]; we do not prove that lemma here. For an object Z of finite length, a linearly independent family of maps $\mathbb{1} \to Z$ gives a monomorphism from the corresponding finite direct sum of copies of $\mathbb{1}$ to Z: any kernel is a direct sum of copies of the simple object $\mathbb{1}$ and would give a linear relation. Thus $\dim_k \mathrm{Hom}(\mathbb{1}, Z) \leq \mathrm{length}(Z)$. Finally rigidity gives

$$\mathrm{Hom}(X, Y) \cong \mathrm{Hom}(\mathbb{1}, X^* \otimes Y),$$

and $X^* \otimes Y$ has finite length by assumption. ∎

15.2 Finite-group actions inside a category

For a finite-dimensional vector space V and $X \in \mathcal{C}$, the notation $V \otimes X$ denotes the corresponding finite direct sum of copies of X, characterized independently of a choice of basis. Define

$$\underline{\mathrm{Hom}}(V, X) = V^* \otimes X.$$

Suppose a finite group G acts on X, that is, we have a homomorphism $G \to \mathrm{Aut}_{\mathcal{C}}(X)$. The averaging operator

$$e_G = \frac{1}{|G|} \sum_{g \in G} g \in \mathrm{End}_{\mathcal{C}}(X)$$

is idempotent. Its image is the invariant object X^G. Its kernel is the sum of the images of $g - 1$, so the coinvariant object $X_G = \mathrm{coker}(\bigoplus_{g \in G} X \xrightarrow{(g-1)} X)$ is canonically isomorphic to $\operatorname{im} e_G$.

If $V \in \mathrm{Rep}_k(G)$, there is a diagonal action on $V^* \otimes X$, and we put

$$\underline{\mathrm{Hom}}_G(V, X) = (V^* \otimes X)^G.$$

This is an object of $\mathcal{C}$, rather than merely the vector space of morphisms from V to X.

15.4 Proposition (Isotypic decomposition)**.** *Let $(V_\lambda)_{\lambda \in I}$ be representatives of the irreducible kG-modules. For every object X carrying a G-action, evaluation induces a natural G-equivariant isomorphism*

$$\bigoplus_{\lambda \in I} V_\lambda \otimes \underline{\mathrm{Hom}}_G(V_\lambda, X) \xrightarrow{\sim} X.$$

Proof. Maschke's theorem and the algebraic closure of k give $kG \cong \prod_{\lambda \in I} \mathrm{End}_k(V_\lambda)$. The central idempotents split X into isotypic summands. In the λ-summand, matrix units split it into $\dim V_\lambda$ isomorphic summands; their multiplicity object is precisely $(V_\lambda^* \otimes X)^G$. The resulting isomorphism is evaluation, so it is independent of the matrix units chosen. ∎

15.3 Partitions and Schur functors

A partition of n is a finite sequence $\lambda = (\lambda_1, \ldots, \lambda_r)$ with $\lambda_1 \geq \cdots \geq \lambda_r > 0$ and $|\lambda| = \sum_i \lambda_i = n$. Its Young diagram is

$$[\lambda] = \{(i, j) : 1 \leq i \leq r,\ 1 \leq j \leq \lambda_i\}.$$

The transpose λ^{t} is defined by interchanging the two coordinates. For example, $(3, 1)^{\mathrm{t}} = (2, 1, 1)$. We also allow the empty partition of 0, with Schur functor equal to the constant unit object; this convention is used in the direct-sum formula below. The irreducible representations of S_n are indexed by partitions of n; write V_λ for the corresponding representation.

The symmetry on $\mathcal{C}$ defines an action of S_n on $X^{\otimes n}$ by adjacent transpositions. The symmetric, rather than merely braided, condition ensures that these transpositions square to the identity.

15.5 Definition. For $\lambda \vdash n$, the *Schur functor* is

$$S^\lambda(X) = \underline{\mathrm{Hom}}_{S_n}(V_\lambda, X^{\otimes n}).$$

Applying the isotypic decomposition gives

$$X^{\otimes n} \cong \bigoplus_{\lambda \vdash n} V_\lambda \otimes S^\lambda(X). \tag{15.1}$$

For the one-row and one-column partitions, $V_{(n)}$ is trivial and $V_{(1^n)}$ is the sign representation. Consequently

$$S^{(n)}(X) = \mathrm{Sym}^n(X), \qquad S^{(1^n)}(X) = \bigwedge^n X.$$

In characteristic zero these are the images of the symmetrizing and antisymmetrizing idempotents, respectively.

15.6 Proposition. *If X is of moderate growth, then some Schur functor annihilates X.*

Proof. Suppose instead that $S^\lambda(X) \neq 0$ for every partition λ. Each summand in (15.1) then contributes at least $\dim V_\lambda$ to the length. Therefore

$$\mathrm{length}(X^{\otimes n}) \geq \sum_{\lambda \vdash n} \dim V_\lambda \geq \left(\sum_{\lambda \vdash n} (\dim V_\lambda)^2 \right)^{1/2} = \sqrt{n!}.$$

The last equality is the dimension of the regular representation of S_n. For even n, $n! \geq (n/2)^{n/2}$, so $(\sqrt{n!})^{1/n} \geq (n/2)^{1/4}$ tends to infinity. This contradicts any exponential bound a_X^n. ∎

15.7 Example. For an ordinary vector space of dimension d, $\bigwedge^{d+1} V = 0$. An odd one-dimensional supervector space instead has $\mathrm{Sym}^2(V) = 0$, while its exterior powers never vanish. Thus exterior powers alone cannot characterize super fiber functors. Schur functors allow both kinds of vanishing.

15.4 Restriction, induction, and multiplicities

For $n = n_1 + \cdots + n_r$, the Young subgroup $S_{n_1} \times \cdots \times S_{n_r} \subseteq S_n$ acts on consecutive blocks of letters. Its irreducibles are the external tensor products $V_{\mu_1} \boxtimes \cdots \boxtimes V_{\mu_r}$, with $\mu_i \vdash n_i$. Define

$$[\lambda : \mu_1, \ldots, \mu_r] = \dim_k \mathrm{Hom}_{\prod_i S_{n_i}} \left(\boxtimes_i V_{\mu_i}, \mathrm{Res}^{S_n}_{\prod_i S_{n_i}} V_\lambda \right).$$

Frobenius reciprocity identifies this with the multiplicity of V_λ in $\mathrm{Ind}^{S_n}_{\prod_i S_{n_i}}(\boxtimes_i V_{\mu_i})$. These nonnegative integers are computed by the Littlewood–Richardson rule, iterated when $r > 2$.

For example, when $r = 2$ the same decomposition yields

$$S^\lambda(X \oplus Y) \cong \bigoplus_{|\mu|+|\nu|=|\lambda|} (S^\mu(X) \otimes S^\nu(Y))^{\oplus [\lambda:\mu,\nu]}.$$

Indeed, expand $(X \oplus Y)^{\otimes n}$ by the subsets of tensor positions occupied by X; each fixed cardinality is induced from its Young subgroup, and then apply the multiplicity-object functor. Likewise,

$$S^\mu(X) \otimes S^\nu(X) \cong \bigoplus_{\lambda \vdash |\mu|+|\nu|} S^\lambda(X)^{\oplus [\lambda:\mu,\nu]}.$$

These formulas explain how the representation theory of symmetric groups controls the operations appearing in the Schur-finiteness criterion.

16 Lecture 16 (February 21, 2025, est.): Negligible morphisms and semisimplification

Scribe: Guillermo Sanmarco

The previous lecture showed that an object on which every Schur functor is nonzero has tensor-power lengths at least $\sqrt{n!}$. This completes the growth argument we will use from the characteristic-zero theory. We now develop a construction that is particularly useful in positive characteristic. Throughout this lecture k is algebraically closed.

16.1 Pseudotensor categories

16.1 Definition. A *symmetric pseudotensor category* over k is an essentially small, additive, k-linear, rigid symmetric monoidal category $\mathcal{C}$ with finite-dimensional morphism spaces and $\mathrm{End}_{\mathcal{C}}(\mathbb{1}) = k$, in which every idempotent splits. The latter condition is also called *Karoubian*: if $e \in \mathrm{End}(X)$ and $e^2 = e$, then $X \cong \mathrm{im}(e) \oplus \mathrm{im}(1-e)$. We do not assume that $\mathcal{C}$ is abelian.

Every additive category has an idempotent completion: its objects are pairs (X, e) with $e^2 = e$, and morphisms $(X, e) \to (Y, f)$ are maps $a : X \to Y$ with $a = fae$. This adds the missing images of idempotents.

16.2 Proposition (Krull–Schmidt)**.** *Every object of $\mathcal{C}$ is a finite direct sum of indecomposable objects, uniquely up to permutation and isomorphism. The endomorphism algebra of an indecomposable object is a finite-dimensional local algebra with residue field k.*

Proof. An orthogonal decomposition of id_X into primitive idempotents in the finite-dimensional algebra $\mathrm{End}(X)$ gives the decomposition of X, because idempotents split. For a primitive idempotent e, the corner algebra $e\,\mathrm{End}(X)e$ is local. Its residue division algebra is finite-dimensional over the algebraically closed field k, hence is k. Uniqueness follows from the usual exchange argument for local endomorphism rings: if id_U is expressed as a sum of composites through the summands of another decomposition, at least one composite is invertible, identifying U with one of those summands; cancel and continue. ∎

In particular, a noninvertible endomorphism of an indecomposable object lies in the radical and is nilpotent. Every endomorphism of such an object is of the form $\lambda\,\mathrm{id} + n$ with $\lambda \in k$ and n nilpotent.

16.2 Tensor ideals and quotient categories

16.3 Definition. A *tensor ideal* $\mathcal{I}$ is a family of linear subspaces $\mathcal{I}(X, Y) \subseteq \mathrm{Hom}_{\mathcal{C}}(X, Y)$ such that

1. if $f \in \mathcal{I}(X, Y)$, $a : W \to X$, and $b : Y \to Z$, then $bfa \in \mathcal{I}(W, Z)$;
2. if $f \in \mathcal{I}(X, Y)$ and $g : U \to V$, then $f \otimes g \in \mathcal{I}(X \otimes U, Y \otimes V)$ and $g \otimes f \in \mathcal{I}(U \otimes X, V \otimes Y)$.

The quotient $\mathcal{C}/\mathcal{I}$ has the same objects and

$$\mathrm{Hom}_{\mathcal{C}/\mathcal{I}}(X,Y) = \mathrm{Hom}_{\mathcal{C}}(X,Y)/\mathcal{I}(X,Y).$$

The ideal conditions make composition and tensor product well defined. The associativity, symmetry, and duality maps descend, giving an additive rigid symmetric monoidal quotient and a quotient functor $Q : \mathcal{C} \to \mathcal{C}/\mathcal{I}$. Every additive symmetric monoidal functor annihilating $\mathcal{I}$ factors uniquely through Q. If an idempotent-complete target is required, one may then take the idempotent completion of the quotient. For the negligible ideal below, the quotient is already semisimple and idempotent complete under the nilpotent-trace hypothesis stated below.

This is a quotient of morphism spaces, not a Serre quotient of an abelian category. In particular, even when both categories happen to be abelian, the quotient functor need not be exact.

16.3 The trace pairing

Recall that for $f : X \to X$, the symmetric categorical trace is

$$\mathrm{Tr}_X(f) = \Big(\mathbb{1} \xrightarrow{\mathrm{coev}_X} X \otimes X^* \xrightarrow{f\otimes \mathrm{id}} X \otimes X^* \xrightarrow{c_{X,X^*}} X^* \otimes X \xrightarrow{\mathrm{ev}_X} \mathbb{1}\Big) \in k.$$

Set $\dim(X) = \mathrm{Tr}_X(\mathrm{id}_X)$. The trace is cyclic, $\mathrm{Tr}_X(gf) = \mathrm{Tr}_Y(fg)$ for $f : X \to Y$, $g : Y \to X$, and is additive on direct sums and multiplicative on tensor products.

16.4 Definition. A morphism $f : X \to Y$ is *negligible* if

$$\mathrm{Tr}_X(gf) = 0 \qquad \text{for every } g : Y \to X.$$

Write $\mathcal{N}(X,Y)$ for the subspace of negligible morphisms. An object is negligible if its identity morphism is negligible.

16.5 Proposition. *The negligible morphisms form a tensor ideal.*

Proof. Linearity is immediate. If $a : W \to X$, $b : Y \to Z$, and $h : Z \to W$, then cyclicity gives $\mathrm{Tr}_W(hbfa) = \mathrm{Tr}_X(ahbf) = 0$, so bfa is negligible. For tensor products, let $f : X \to Y$ be negligible, $g : U \to V$, and $h : Y \otimes V \to X \otimes U$. Close the U-strand of $h(\mathrm{id}_Y \otimes g)$ using evaluation, coevaluation and symmetry; this defines a partial trace $h' : Y \to X$. The trace identities give

$$\mathrm{Tr}_{X\otimes U}\big(h(f \otimes g)\big) = \mathrm{Tr}_X(h'f) = 0.$$

Thus $f \otimes g$ is negligible. Symmetry gives the assertion for $g \otimes f$ as well. ∎

16.4 Benson's lemma

From now on assume additionally that

$$\mathrm{Tr}_X(n) = 0 \quad \text{for every nilpotent } n \in \mathrm{End}_{\mathcal{C}}(X). \tag{16.1}$$

This hypothesis is important for a pseudotensor category. It holds for the representation categories used below, where the categorical trace is the ordinary trace. We use the following criterion, often called Benson's lemma; see also [EO19, §2].

16.6 Lemma. *Let X, Y be nonzero indecomposable objects. A map $f : X \to Y$ is negligible if and only if* $\dim(Y) = 0$ *or f is not an isomorphism. For decompositions $X = \bigoplus_i X_i$ and $Y = \bigoplus_j Y_j$ into indecomposables, a map $f = (f_{ji})$ is negligible if and only if each component $f_{ji} : X_i \to Y_j$ is negligible.*

Proof. Suppose $f : X \to Y$ is not invertible. For any $g : Y \to X$, the endomorphism fg cannot be invertible: otherwise, with $h = (fg)^{-1}$, the map $s = gh : Y \to X$ satisfies $fs = \mathrm{id}_Y$. The nonzero idempotent sf splits X as $Y \oplus Z$. Since X is indecomposable, $Z = 0$, which would make f invertible. Thus fg is a nonunit of the local algebra $\mathrm{End}(Y)$, hence is nilpotent. By (16.1), $\mathrm{Tr}_Y(fg) = 0$.

If f is invertible and $\dim(Y) \neq 0$, taking $g = f^{-1}$ proves that f is not negligible. If $\dim(Y) = 0$, write $fg = \lambda \,\mathrm{id}_Y + n$ with n nilpotent. Then $\mathrm{Tr}_Y(fg) = \lambda \dim(Y) + \mathrm{Tr}_Y(n) = 0$, so f is negligible.

For the last assertion, if all components are negligible, then for $g = (g_{ij}) : Y \to X$,

$$\mathrm{Tr}_Y(fg) = \sum_{i,j} \mathrm{Tr}_{Y_j}(f_{ji} g_{ij}) = 0.$$

Conversely, test against matrices g with just one nonzero component. This isolates each trace pairing and proves that every f_{ji} is negligible. ∎

16.7 Theorem. *The quotient $\overline{\mathcal{C}} = \mathcal{C}/\mathcal{N}$ is a semisimple symmetric tensor category. Its simple objects are the images of the indecomposable objects of $\mathcal{C}$ of nonzero categorical dimension, one for each isomorphism class. An object becomes zero if and only if every indecomposable summand has dimension zero.*

Proof. For nonisomorphic indecomposables X, Y, every map $X \to Y$ is negligible. If $\dim(X) = 0$, the identity of X is negligible, so $Q(X) = 0$. Otherwise Lemma 16.6 identifies the negligible endomorphisms of X with the radical of $\mathrm{End}(X)$, and

$$\mathrm{End}_{\overline{\mathcal{C}}}(Q(X)) = \mathrm{End}_{\mathcal{C}}(X)/\operatorname{rad}\mathrm{End}_{\mathcal{C}}(X) \cong k.$$

Every object is a finite direct sum of these images. Hence the quotient is equivalent, as an additive category, to the category of finite-support families of finite-dimensional vector spaces indexed by the surviving indecomposables. This category is abelian, semisimple and idempotent complete. The quotient retains rigidity and symmetry, and the unit survives because $\dim(\mathbb{1}) = 1$. ∎

16.8 Definition. The category $\overline{\mathcal{C}}$ is the *semisimplification* of $\mathcal{C}$, and Q is its semisimplification functor.

An object of total categorical dimension zero need not become zero: its nonzero-dimensional indecomposable summands may have dimensions whose sum is zero in k. The criterion in the theorem is a condition on every indecomposable summand, not only on their sum.

17 Lecture 17 (February 24, 2025, est.): The Verlinde category

Scribe: Guillermo Sanmarco

Throughout this lecture, k is algebraically closed of characteristic $p > 0$. We apply semisimplification to representations of the cyclic group C_p. Recall that an indecomposable survives semisimplification precisely when its categorical dimension is nonzero, provided traces of nilpotent endomorphisms vanish. This hypothesis holds in representation categories, where the categorical trace is the ordinary linear trace.

17.1 Representations of the cyclic group

Choose a generator σ of C_p. Writing $t = \sigma - 1$ gives

$$k[C_p] = k[\sigma]/(\sigma^p - 1) \cong k[t]/(t^p).$$

Thus a representation is a vector space with a nilpotent endomorphism whose pth power is zero. Jordan normal form gives exactly p indecomposable representations:

$$J_n = k[t]/(t^n), \qquad 1 \le n \le p.$$

In particular, J_1 is the trivial representation and J_p is the regular representation.

17.1 Definition. The *Verlinde category* is the symmetric tensor category

$$\mathrm{Ver}_p = \overline{\mathrm{Rep}_k(C_p)}.$$

Write L_n for the image of J_n in this semisimplification. Its simple objects are $L_1, \dots, L_{p-1}$, with $L_1 = \mathbb{1}$, whereas $L_p = 0$. Their categorical dimensions are $\dim(L_n) = n \cdot 1_k$.

There are finitely many simples here, so Ver_p is a symmetric fusion category. Finiteness of the original group alone would not ensure this conclusion for its semisimplification: representation categories of other finite groups can have infinitely many surviving indecomposables. See [EO19] for examples and the general construction.

17.2 Tensor products of Jordan blocks

17.2 Proposition. *The J_n are self-dual, and*

$$J_2 \otimes J_n \cong \begin{cases} J_2, & n = 1, \\ J_{n-1} \oplus J_{n+1}, & 2 \le n \le p-1, \\ J_p^{\oplus 2}, & n = p. \end{cases}$$

For $1 \le n \le p-1$ one also has

$$J_{p-1} \otimes J_n \cong J_{p-n} \oplus J_p^{\oplus(n-1)}.$$

Proof. On a dual representation, σ acts by the inverse transpose. The inverse of a unipotent Jordan block is again a unipotent matrix with one Jordan block of the same size, proving self-duality.

For the first tensor-product formula, identify the underlying vector space with $k[x,y]/(x^2,y^n)$. The generator acts by multiplication by $(1+x)(1+y)$. Set $u = x(1+y)$; this is an invertible change of variable, so the algebra is also $k[u,y]/(u^2,y^n)$, and t acts by multiplication by $u+y$. Hence

$$t^r = y^r + ruy^{r-1}.$$

For $2 \le n < p$, the nilpotency index is $n+1$, since $t^n = nuy^{n-1} \neq 0$. Moreover, the kernel of multiplication by t has dimension two. Indeed, $t(A(y)+uB(y)) = 0$ means

$$yA = 0, \qquad A + yB = 0,$$

so $A = ay^{n-1}$ and $B = -ay^{n-2} + by^{n-1}$. There are therefore two Jordan blocks; their sizes are $n+1$ and $n-1$ because the total dimension is $2n$. If $n = p$, the same kernel computation gives two blocks, each of size at most p, and their total size is $2p$. The case $n = 1$ is immediate.

Finally, tensor the exact sequence

$$0 \longrightarrow J_{p-1} \longrightarrow J_p \longrightarrow J_1 \longrightarrow 0$$

with J_n. The middle term is $J_p^{\oplus n}$: for any finite group, the regular representation tensored diagonally with V is isomorphic to $\dim(V)$ copies of the regular representation, by the map $g \otimes v \mapsto g \otimes g^{-1}v$. A surjection $(k[t]/t^p)^n \to k[t]/t^n$ can be changed by an invertible basis transformation of its source to the standard quotient on its first factor and zero on the others. To see this, one of the images of the standard generators must have nonzero constant term; make it the first generator, multiply it by a unit to make its image 1, and subtract suitable multiples of it from the remaining generators. The kernel is then $t^n(k[t]/t^p) \oplus (k[t]/t^p)^{n-1}$, as claimed. ∎

17.3 Theorem (Fusion rules)**.** *For $1 \le i,j \le p-1$,*

$$L_i \otimes L_j \cong \bigoplus_{r=1}^{\min(i,j,p-i,p-j)} L_{|i-j|+2r-1}.$$

In particular,

$$L_2 \otimes L_n = L_{n-1} \oplus L_{n+1}, \qquad L_{p-1} \otimes L_n = L_{p-n},$$

where $L_0 = L_p = 0$ in the first formula.

Proof. The two displayed special cases follow from the proposition. For $p > 2$, write $x = [L_2]$ in the Grothendieck ring and define polynomials $f_0 = 0$, $f_1 = 1$, $f_{n+1} = xf_n - f_{n-1}$. The recurrence gives $[L_n] = f_n(x)$ and $f_p(x) = 0$. Induction using the same recurrence gives the polynomial identity

$$f_i(x)f_j(x) = \sum_{r=1}^{\min(i,j)} f_{|i-j|+2r-1}(x).$$

Modulo $f_p(x)$ one has $f_{p+a}(x) = -f_{p-a}(x)$ for $a \geq 0$ in the range in question. The terms above p consequently cancel the reflected terms below p, and the term f_p vanishes. The remaining indices start at $|i-j|+1$ and end at $\min(i+j-1, 2p-i-j-1)$, giving the formula. Since the category is semisimple, equality of these classes is an isomorphism of objects. For $p = 2$ there is only the simple unit, and the formula is immediate. ∎

These are the truncated Clebsch–Gordan rules also occurring in the semisimplified quantum $\mathfrak{sl}_2$ category at a primitive $2p$th root of unity in characteristic zero. This comparison concerns the fusion rules; the categories have different ground-field characteristics. For the positive-characteristic construction and fusion rules, see [Ost15, Section 3].

17.3 Small characteristics and failure of fiber functors

For $p = 2$, the only surviving simple is the unit, so $\mathrm{Ver}_2 \simeq \mathrm{Vect}_k$. For $p = 3$, the simples are L_1 and L_2, with $L_2^{\otimes 2} = \mathbb{1}$. Its self-symmetry is -1: in $J_2 \otimes J_2$, the alternating line is a copy of J_1 on which the flip is -1, whereas the symmetric square is J_3 and disappears. Thus $\mathrm{Ver}_3 \simeq \mathrm{SVec}_k$.

For $p = 5$, the fusion rule gives

$$L_3 \otimes L_3 \cong \mathbb{1} \oplus L_3.$$

If there were a symmetric fiber functor to SVec_k, the total vector space dimension n of the image of L_3 would be a positive integer satisfying $n^2 = n + 1$, which is impossible. Here one must use an ordinary nonnegative integer dimension, not the categorical dimension in k: the categorical dimension $3 \in k$ does satisfy this equation in characteristic 5.

17.4 Proposition. *For every prime $p \geq 5$, Ver_p has no fiber functor to Vect_k or to SVec_k.*

Proof. Suppose such a functor exists, and let d_i be the total dimension of the image of L_i. Faithfulness gives positive integers d_i for $1 \leq i \leq p-1$. Set $d_0 = d_p = 0$, $d_1 = 1$, and $d = d_2$. The fusion rules imply

$$d d_i = d_{i-1} + d_{i+1} \quad (1 \leq i \leq p-1).$$

Since $d^2 = 1 + d_3 \geq 2$, one has $d \geq 2$. Starting with $d_1 - d_0 = 1$, the recurrence gives

$$d_{i+1} - d_i = (d-2)d_i + (d_i - d_{i-1}) \geq 1$$

by induction, including the last step $i = p-1$. Thus $d_p \geq p$, contrary to $d_p = 0$. ∎

17.5 Example (A characteristic-two obstruction)**.** There are also symmetric tensor categories in characteristic 2 with no fiber functor to Vect_k, even though Ver_2 itself is Vect_k. Consider the Hopf algebra

$$H = k[d]/(d^2), \qquad \Delta(d) = d \otimes 1 + 1 \otimes d,$$

with triangular structure $R = 1 \otimes 1 + d \otimes d$. The identities $(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23}$ and $(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12}$ follow by expanding and using $d^2 = 0$; also $R_{21}R = 1$ because $2 = 0$. Thus the symmetry on $\mathrm{Rep}(H)$ is $c = \tau \circ R$, where τ is the usual flip.

Let X have basis e, f with $de = 0$ and $df = e$. Then

$$c(e \otimes e) = e \otimes e, \qquad c(e \otimes f) = f \otimes e, \qquad c(f \otimes f) = f \otimes f + e \otimes e.$$

The image of $\mathrm{id} - c$ is spanned by $e \otimes e$ and $e \otimes f + f \otimes e$. Consequently $\mathrm{Sym}^2 X = \mathrm{coker}(\mathrm{id} - c)$ is two-dimensional, with basis the classes of $e \otimes f$ and $f \otimes f$. The action of d on both classes is zero, so $\mathrm{Sym}^2 X \cong \mathbb{1}^{\oplus 2}$. On the other hand, X is an extension of $\mathbb{1}$ by $\mathbb{1}$. An exact symmetric fiber functor F to Vect_k would give $\dim F(X) = 2$ and preserve this cokernel. This would imply both $\dim F(\mathrm{Sym}^2 X) = 2$ and $\dim \mathrm{Sym}^2(F(X)) = 3$, a contradiction.

17.4 Tensor subcategories of the Verlinde category

Assume now that $p \geq 5$. Let Ver_p^+ be the full semisimple subcategory with simples

$$L_1, L_3, L_5, \ldots, L_{p-2}.$$

The fusion rules show that it is closed under tensor products and duals, and that it is generated by L_3. More explicitly, $L_3 \otimes L_{2r+1}$ contains L_{2r+3} whenever $3 \leq 2r + 3 \leq p - 2$, starting with L_3 itself.

The object $\varepsilon = L_{p-1}$ is invertible of order two, with $\dim(\varepsilon) = -1$. For an invertible object in a symmetric category, the self-symmetry is its categorical dimension times the identity (compute the trace using its inverse object). Thus the subcategory generated by ε is SVec_k, with ε corresponding to the odd line.

17.6 Proposition. *The full tensor subcategories of* Ver_p *are exactly*

$$\mathrm{Vect}_k, \qquad \mathrm{SVec}_k, \qquad \mathrm{Ver}_p^+, \qquad \mathrm{Ver}_p\,.$$

Moreover, multiplication induces a symmetric equivalence

$$\mathrm{SVec}_k \boxtimes \mathrm{Ver}_p^+ \xrightarrow{\sim} \mathrm{Ver}_p\,.$$

Here a full tensor subcategory is understood to be closed under finite direct sums and subobjects, and $\boxtimes$ *denotes the product whose simple objects are pairs of simple objects of its factors.*

Proof. If a tensor subcategory contains L_i with $2 \leq i \leq p - 2$, then $L_i \otimes L_i$ contains L_3, so the subcategory contains Ver_p^+. If i is even, then $p - i$ is odd, hence L_{p-i} already belongs to Ver_p^+. The product $L_i \otimes L_{p-i}$ contains $L_{p-1} = \varepsilon$, so the subcategory is all of Ver_p. More generally, adjoining ε to Ver_p^+ gives all the even-indexed simples by $\varepsilon \otimes L_j = L_{p-j}$ for odd j. If there are no such intermediate simples, the only possibilities are Vect_k and the subcategory generated by ε. This proves the list.

The multiplication functor in the displayed equivalence sends $\mathbb{1}\boxtimes L_j$ to L_j and $\varepsilon\boxtimes L_j$ to L_{p-j} for odd j. These are all the simple objects of Ver_p, each appearing once. It identifies the corresponding morphism spaces, and is therefore an equivalence of semisimple categories. The symmetry allows the middle two factors of a tensor product to be interchanged, supplying its symmetric monoidal structure. ■

This decomposition and the list of subcategories are also proved in [Ost15, Proposition 3.3]. The product construction and its universal property will be made explicit in the next lecture.

18 Lecture 18 (February 26, 2025, est.): Products, equivariant objects, and tensor powers

Scribe: Guillermo Sanmarco

Throughout this lecture, k is algebraically closed. Our aim is to construct the Frobenius functor for a semisimple symmetric tensor category in characteristic p. We first make precise the product of categories and the category of objects equipped with a group action. These constructions will also explain the product decomposition of Ver_p from the previous lecture.

18.1 Products of linear categories

Let $\mathcal{C}$ and $\mathcal{D}$ be k-linear additive Karoubian categories. Begin with formal pairs $X \boxtimes Y$ and put

$$\mathrm{Hom}(X \boxtimes Y, X' \boxtimes Y') = \mathrm{Hom}_{\mathcal{C}}(X, X') \otimes_k \mathrm{Hom}_{\mathcal{D}}(Y, Y').$$

Composition is componentwise on elementary tensors. One must then take the additive hull, allowing finite formal direct sums, and its idempotent completion. We denote the resulting Karoubian product by $\mathcal{C} \boxtimes \mathcal{D}$ in this lecture. Merely taking pairs would not in general give an additive category.

If the factors are rigid symmetric monoidal, the formulas

$$(X \boxtimes Y) \otimes (X' \boxtimes Y') = (X \otimes X') \boxtimes (Y \otimes Y'), \qquad (X \boxtimes Y)^* = X^* \boxtimes Y^*$$

extend to give the product the same structures. There are canonical functors $X \mapsto X \boxtimes \mathbb{1}$ and $Y \mapsto \mathbb{1} \boxtimes Y$.

18.1 Proposition (Universal property)**.** *For a k-linear additive Karoubian symmetric monoidal category $\mathcal{A}$, giving an additive k-linear symmetric monoidal functor $\mathcal{C} \boxtimes \mathcal{D} \to \mathcal{A}$ is equivalent to giving one such functor from each factor to $\mathcal{A}$.*

Proof. Given $F : \mathcal{C} \to \mathcal{A}$ and $G : \mathcal{D} \to \mathcal{A}$, define $H(X \boxtimes Y) = F(X) \otimes G(Y)$ and $H(f \otimes g) = F(f) \otimes G(g)$. Its tensor constraint uses the symmetry to interchange the middle two factors:

$$F(X) \otimes G(Y) \otimes F(X') \otimes G(Y') \longrightarrow F(X \otimes X') \otimes G(Y \otimes Y').$$

The symmetry axioms give the required coherence. Additivity and splitting idempotents extend H to the completed product, uniquely up to natural isomorphism. Conversely, restrict a functor H to the two canonical copies of the factors. Since $X \boxtimes Y = (X \boxtimes \mathbb{1}) \otimes (\mathbb{1} \boxtimes Y)$, the two procedures are inverse, including on monoidal natural isomorphisms. ∎

18.2 Remark. This Karoubian product should not be confused with the general abelian Deligne tensor product, characterized by right-exact bilinear functors. The distinction disappears in the case used here, where one factor is split semisimple

and the other is abelian. If $\mathcal{C}$ is split semisimple with simple objects S, then, as a linear category,

$$\mathcal{C} \boxtimes \mathcal{D} \simeq \bigoplus_{S \in \mathrm{Irr}(\mathcal{C})} \mathcal{D},$$

where objects have finite support. Indeed, every object of the product is a finite sum of objects $S \boxtimes Y_S$, there are no morphisms between distinct S-blocks, and each block has exactly the morphisms of $\mathcal{D}$. Splitting idempotents preserves this description. In particular the product is abelian if $\mathcal{D}$ is. When both factors are split semisimple, its simples are precisely the pairs of simples.

18.3 Proposition. *Let $\mathcal{A}$ and $\mathcal{B}$ be full semisimple tensor subcategories of a semisimple symmetric tensor category $\mathcal{C}$. If their only common simple object is $\mathbb{1}$, then multiplication gives a symmetric equivalence*

$$\mathcal{A} \boxtimes \mathcal{B} \simeq \langle \mathcal{A}, \mathcal{B} \rangle,$$

where the right-hand side is the full tensor subcategory generated by the two subcategories.

Proof. For simple $X, X' \in \mathcal{A}$ and $Y, Y' \in \mathcal{B}$, rigidity and symmetry identify

$$\mathrm{Hom}(X \otimes Y, X' \otimes Y') \cong \mathrm{Hom}(X'^* \otimes X, Y' \otimes Y^*).$$

An object of $\mathcal{A}$ and an object of $\mathcal{B}$ can have only copies of the unit as common simple summands. The last Hom space is therefore the tensor product of the corresponding unit multiplicity spaces. Its dimension is 1 if $X \cong X'$ and $Y \cong Y'$, and is 0 otherwise. In particular, the $X \otimes Y$ are simple and distinct for distinct pairs. This proves full faithfulness of multiplication on the product, and its image is closed under sums and summands. Every tensor word in the two subcategories can be reordered to have all its $\mathcal{A}$ factors first, so the image is exactly the generated subcategory. The symmetry gives the monoidal structure as in the universal property above. ∎

For example, the subcategories SVec_k and Ver_p^+ inside Ver_p, for $p \geq 5$, have only the simple unit in common and generate the whole category. The proposition recovers their product decomposition.

18.2 Equivariant objects

An action of a group G on a category requires coherent functors, not just a homomorphism into the group of isomorphism classes of autoequivalences. Concretely, choose autoequivalences T_g and natural isomorphisms

$$\mu_{g,h} : T_g T_h \xrightarrow{\sim} T_{gh},$$

with a unit identification for T_1, satisfying the associativity cocycle identity. For a monoidal action, the T_g and the isomorphisms are monoidal.

18.4 Definition. An *equivariant object* is an object X together with isomorphisms $u_g : T_g(X) \to X$ satisfying

$$u_{gh} \circ \mu_{g,h,X} = u_g \circ T_g(u_h)$$

and the unit condition. A morphism $f : (X, u) \to (Y, v)$ satisfies $f \circ u_g = v_g \circ T_g(f)$ for all g. The resulting category is denoted $\mathcal{C}^G$.

For a monoidal action, tensor products of the maps u_g give $\mathcal{C}^G$ a monoidal structure. If the action respects a symmetry, the symmetry is equivariant as well. In the rest of this lecture the action is *trivial*: $T_g = \mathrm{id}$ and all coherence maps are identities. Then an equivariant object is simply (X, ρ) with $\rho : G \to \mathrm{Aut}_{\mathcal{C}}(X)$ a group homomorphism. For example, $\mathrm{Vect}_k^G = \mathrm{Rep}_k(G)$.

18.5 Proposition. *Let $\mathcal{C}$ be a semisimple symmetric tensor category and G a finite group acting trivially. There is a symmetric equivalence*

$$\mathcal{C}^G \simeq \mathcal{C} \boxtimes \mathrm{Rep}_k(G).$$

Proof. Write an object of $\mathcal{C}$ as $X = \bigoplus_S S \otimes_k M_S$, where the S are simple and the $M_S = \mathrm{Hom}(S, X)$ are finite-dimensional multiplicity spaces. The notation $S \otimes_k M_S$ means a direct sum of copies of S. Since $\mathrm{End}(S) = k$ and morphisms between distinct simples vanish, a G-action on X is exactly a G-representation on each M_S. Equivariant morphisms are exactly the equivariant linear maps on these spaces. This gives the claimed equivalence of categories. It is the functor sending $X \boxtimes V$ to $X \otimes_k V$ with G acting on V, and the componentwise tensor actions show that it is symmetric monoidal. ∎

18.3 Semisimplifying equivariant objects

18.6 Lemma. *A simple object S of a semisimple rigid symmetric category has nonzero categorical dimension.*

Proof. Rigidity gives $\dim_k \mathrm{Hom}(\mathbb{1}, S \otimes S^*) = \dim_k \mathrm{End}(S) = 1$, so $\mathbb{1}$ occurs with multiplicity one in $S \otimes S^*$. Both coevaluation and evaluation composed with the symmetry are nonzero, by the duality identities. In a semisimple category, a nonzero inclusion and a nonzero projection of a simple summand of multiplicity one have nonzero composite. This composite is $\dim(S)$. ∎

This assertion concerns simple objects. It does not say that all nonzero objects have nonzero dimension: in characteristic p, $\mathbb{1}^{\oplus p}$ has dimension zero.

18.7 Proposition. *In the situation of Proposition 18.5,*

$$\overline{\mathcal{C}^G} \simeq \mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}$$

as symmetric tensor categories.

Proof. The indecomposable objects of $\mathcal{C}^G$ are $S \boxtimes V$, where S is simple in $\mathcal{C}$ and V is an *indecomposable* representation of G; V need not be simple. The trace of an endomorphism of such an object is $\dim(S)$ times its ordinary trace on V. Thus nilpotent endomorphisms have trace zero, and

$$\dim(S \boxtimes V) = \dim(S)\dim_k(V).$$

By the lemma, this vanishes precisely when $\dim_k(V)$ vanishes in k. More generally, on each S-block the trace pairing on morphism spaces is the ordinary trace pairing on G-modules multiplied by the nonzero scalar $\dim(S)$. Its radical is therefore unchanged. Quotienting by these radicals gives exactly $\mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}$. Tensor products and symmetries descend from the equivalence before quotienting. ∎

18.4 The cyclic action on a tensor power

Assume now that $\operatorname{char} k = p > 0$ and that $\mathcal{C}$ is semisimple. The symmetry supplies an S_p-action on $X^{\otimes p}$. Restricting to the cycle $\sigma = (1\,2\,\cdots\,p)$ defines a functor

$$P : \mathcal{C} \longrightarrow \mathcal{C}^{C_p}, \qquad X \longmapsto (X^{\otimes p}, \sigma), \qquad f \longmapsto f^{\otimes p}.$$

Canonical rearrangement of tensor factors makes P symmetric monoidal. It is not additive. Compose it with semisimplification:

$$\mathrm{Fr}_0 : \mathcal{C} \xrightarrow{P} \mathcal{C}^{C_p} \longrightarrow \overline{\mathcal{C}^{C_p}} \simeq \mathcal{C} \boxtimes \mathrm{Ver}_p\,.$$

The essential point is that the mixed terms in a tensor power become negligible. In the expansion of $(X \oplus Y)^{\otimes p}$, the summands are indexed by words of length p in the letters X, Y. There are two constant words, giving $X^{\otimes p}$ and $Y^{\otimes p}$. Every other word has a free orbit under cyclic rotation because p is prime. Choose a summand W in such an orbit. Using the cyclic action to identify its translates, the sum of the orbit is the equivariant object $W \otimes_k k[C_p]$. Under the product description above, it is $W \boxtimes J_p$ and vanishes after semisimplification. Consequently the canonical inclusions and projections induce

$$\mathrm{Fr}_0(X \oplus Y) \cong \mathrm{Fr}_0(X) \oplus \mathrm{Fr}_0(Y).$$

The same orbit argument applied to morphisms proves additivity; we give that proof in the next lecture. Scalars satisfy

$$\mathrm{Fr}_0(\lambda f) = \lambda^p \mathrm{Fr}_0(f).$$

To express this as a k-linear functor, let $\mathcal{C}^{(1)}$ have the same underlying additive symmetric category as $\mathcal{C}$, but scalar action $\lambda \cdot_{(1)} f = \lambda^p f$. The field k is perfect, so this is again a category over k. Twist the whole target of Fr_0 in this manner, and use the fact that Ver_p is defined over $\mathbb{F}_p$ to identify its twist with itself. The resulting functor is

$$\mathrm{Fr} : \mathcal{C} \longrightarrow \mathcal{C}^{(1)} \boxtimes \mathrm{Ver}_p\,.$$

This is the Frobenius functor of [Ost15, Section 3.3]. After additivity is established, it is exact because every exact sequence in the source splits.

18.8 Remark. Semisimplicity is essential to the product description used in this construction. For a general symmetric tensor category one cannot obtain the correct Frobenius functor simply by taking the ordinary semisimplification of $\mathcal{C}^{C_p}$. The later lectures use a relative construction to treat that case. The questions of when a category admits a fiber functor to Ver_p, and why Ver_p cannot be further compressed by a tensor functor, will be addressed after studying this Frobenius functor.

19 Lecture 19 (February 28, 2025, est.): The Frobenius functor

Scribe: Guillermo Sanmarco

Throughout this lecture, k is algebraically closed of characteristic $p > 0$, and $\mathcal{C}$ is a semisimple symmetric tensor category. We write $\overline{\mathcal{A}}$ for the semisimplification of a symmetric category $\mathcal{A}$ by negligible morphisms. For the cyclic group C_p, let J_r be the indecomposable representation of dimension r, and let L_r be its image in Ver_p. Thus $L_p = 0$ and $L_1 = \mathbf{1}$.

19.1 Equivariant objects and semisimplification

If a finite group G acts trivially on $\mathcal{C}$, an object of $\mathcal{C}^G$ is a pair (X, ρ) with $\rho : G \to \mathrm{Aut}_{\mathcal{C}}(X)$. The morphisms commute with the specified actions. There is a symmetric equivalence

$$\mathcal{C}^G \simeq \mathcal{C} \boxtimes \mathrm{Rep}_k(G).$$

Indeed, write $X = \bigoplus_S S \otimes_k M_S$, where S runs through the simple objects of $\mathcal{C}$ and $M_S = \mathrm{Hom}(S, X)$. Since $\mathrm{End}(S) = k$ and there are no morphisms between distinct simples, the G-action is precisely a representation on each multiplicity space M_S.

19.1 Proposition. *There is a symmetric equivalence*

$$\overline{\mathcal{C}^G} \simeq \mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}.$$

Proof. The indecomposables of $\mathcal{C}^G$ are $S \boxtimes M$, with S simple and M an *indecomposable* G-module. Their categorical dimensions are $\dim(S)\dim_k(M)$. A simple object in a semisimple rigid category has nonzero categorical dimension: the two one-dimensional spaces $\mathrm{Hom}(\mathbf{1}, S \otimes S^*)$ and $\mathrm{Hom}(S \otimes S^*, \mathbf{1})$ pair nondegenerately, because the unit occurs with multiplicity one. Consequently $S \boxtimes M$ has dimension zero exactly when M does. The trace pairing on the S-isotypic block is the ordinary trace pairing on the multiplicity spaces multiplied by the nonzero scalar $\dim(S)$. Thus taking its radical, and then the quotient, acts separately on each G-module block. This gives the claimed equivalence, compatibly with tensor products and symmetry. ∎

19.2 Remark. It is simple objects, rather than all nonzero objects, that have nonzero dimension here. For example, $\mathbf{1}^{\oplus p}$ is nonzero but has dimension zero in characteristic p.

19.2 Tensor powers and additivity

The symmetry defines an action of S_p on $X^{\otimes p}$. Restrict it to $C_p = \langle \sigma \rangle$, where $\sigma = (1\,2\,\cdots\,p)$. This gives a functor

$$P : \mathcal{C} \longrightarrow \mathcal{C}^{C_p}, \qquad X \longmapsto (X^{\otimes p}, \sigma), \quad f \longmapsto f^{\otimes p}.$$

The canonical regrouping of tensor factors makes P symmetric monoidal, but P is not additive. Compose it with semisimplification to obtain

$$\mathrm{Fr}_0 : \mathcal{C} \longrightarrow \overline{\mathcal{C}^{C_p}} \simeq \mathcal{C} \boxtimes \mathsf{Ver}_p.$$

19.3 Lemma. *The functor* Fr_0 *is additive.*

Proof. For $f, g : X \to Y$, expand $(f+g)^{\otimes p}$. Besides $f^{\otimes p}$ and $g^{\otimes p}$, every term is a word in f and g containing both letters. The cyclic group acts freely on these words, since p is prime. The sum over one orbit has the form

$$T = \sum_{i=0}^{p-1} \sigma_Y^i h \sigma_X^{-i}, \qquad h = h_1 \otimes \cdots \otimes h_p.$$

It is C_p-equivariant. For an equivariant map $u : Y^{\otimes p} \to X^{\otimes p}$, naturality and cyclicity of the trace give

$$\mathrm{Tr}(\sigma_Y^i h \sigma_X^{-i} u) = \mathrm{Tr}(h \sigma_X^{-i} u \sigma_Y^i) = \mathrm{Tr}(hu).$$

Hence $\mathrm{Tr}(Tu) = p\,\mathrm{Tr}(hu) = 0$. Each orbit sum is negligible, so the mixed terms vanish after semisimplification. This proves additivity on morphisms. Since $P(0) = 0$, preservation of biproducts follows as well. ∎

19.3 The scalar twist

Although additive, Fr_0 satisfies $\mathrm{Fr}_0(\lambda f) = \lambda^p \mathrm{Fr}_0(f)$. Define $\mathcal{C}^{(1)}$ to have the same underlying additive symmetric monoidal category as $\mathcal{C}$, with scalar multiplication

$$\lambda \cdot_{(1)} f = \lambda^p f.$$

Because k is perfect, this is again a tensor category over k. The category Ver_p is defined over $\mathbb{F}_p$, and therefore $\mathsf{Ver}_p^{(1)} \simeq \mathsf{Ver}_p$.

19.4 Definition. The *Frobenius functor* is the resulting k-linear symmetric monoidal functor

$$\mathrm{Fr} : \mathcal{C} \longrightarrow \mathcal{C}^{(1)} \boxtimes \mathsf{Ver}_p.$$

It is exact because every exact sequence in $\mathcal{C}$ splits. This argument uses semisimplicity; exactness will be a genuine condition in the last lecture. The construction and additivity argument are due to Ostrik [Ost15, Section 3.3].

19.4 The example Ver_5^+

Let $X = L_3$ in Ver_5^+. Its fusion rule is $X^{\otimes 2} = \mathbf{1} \oplus X$, so

$$X^{\otimes 5} = \mathbf{1}^{\oplus 3} \oplus X^{\oplus 5}.$$

As an equivariant object this has the form $\mathbf{1} \boxtimes M_1 \oplus X \boxtimes M_X$, with $\dim_k M_1 = 3$ and $\dim_k M_X = 5$. Put $\varphi = (1+\sqrt{5})/2 = \mathrm{FPdim}(X)$. Tensor functors between fusion categories preserve Frobenius–Perron dimensions, hence

$$\varphi = \mathrm{FPdim}(\overline{M_1}) + \varphi\,\mathrm{FPdim}(\overline{M_X}).$$

The possible decompositions of a three-dimensional C_5-module are J_3, $J_2 \oplus J_1$, and $J_1^{\oplus 3}$. Their semisimplifications have Frobenius–Perron dimensions φ, $\varphi + 1$, and 3, respectively. Thus equality forces $M_1 = J_3$ and $\overline{M_X} = 0$. Since M_X has dimension five, it must be J_5. Therefore

$$\mathrm{Fr}(X) = \mathbf{1} \boxtimes L_3.$$

19.5 The induced operation on the Grothendieck ring

Write $K_0(\mathcal{C})$ for the Grothendieck ring; a Frobenius twist does not change this ring. Categorical dimension defines a ring map

$$\phi : K_0(\mathcal{C}) \otimes_{\mathbb{Z}} K_0(\mathsf{Ver}_p) \longrightarrow K_0(\mathcal{C}) \otimes_{\mathbb{Z}} k, \qquad [Y] \otimes [L_r] \longmapsto [Y] \otimes r.$$

19.5 Lemma. *For every $X \in \mathcal{C}$, one has $\phi([\mathrm{Fr}(X)]) = [X]^p$.*

Proof. Decompose $P(X)$ as $\bigoplus_S S \boxtimes M_S$. Forgetting the cyclic action gives $[X]^p = \sum_S \dim_k(M_S)[S]$. On semisimplifying, each J_r with $r < p$ contributes categorical dimension r, while J_p is killed and its dimension is already zero in k. The two expressions agree. ∎

20 Lecture 20 (March 3, 2025, est.): Frobenius type and lifting

Scribe: Guillermo Sanmarco

We continue over an algebraically closed field k of characteristic $p > 0$. Our categories are semisimple unless otherwise specified. The Frobenius functor takes values in $\mathcal{C}^{(1)} \boxtimes \mathsf{Ver}_p$, with the scalar twist of Lecture 19.

20.1 Frobenius on the Verlinde category

Assume $p > 2$, and put $\varepsilon = L_{p-1}$. Recall the fusion rules

$$L_r \otimes L_s = \bigoplus_{i=1}^{\min(r,s,p-r,p-s)} L_{|r-s|+2i-1}, \qquad \varepsilon \otimes L_s = L_{p-s}.$$

In particular $\varepsilon^{\otimes 2} = \mathbf{1}$, and $\langle \varepsilon \rangle \simeq \mathrm{SVec}$. The subcategory Ver_p^+ consists of sums of the L_s with s odd.

20.1 Proposition. *Using the prime-field identification* $\mathsf{Ver}_p^{(1)} \simeq \mathsf{Ver}_p$*, one has*

$$\mathrm{Fr}(L_s) = \begin{cases} \mathbf{1} \boxtimes L_s, & s \text{ odd}, \\ \varepsilon \boxtimes L_{p-s}, & s \text{ even}. \end{cases}$$

Proof. Set $x = [L_2]$ in $K_0(\mathsf{Ver}_p)$. The recurrence $[L_{s+1}] = x[L_s] - [L_{s-1}]$, together with $[L_0] = [L_p] = 0$, gives

$$x^p = -2[\varepsilon] \quad \text{in } K_0(\mathsf{Ver}_p) \otimes k.$$

One way to verify this polynomial identity is to put $[L_s] = U_{s-1}(x/2)$, where $U_0 = 1$, $U_1 = 2t$, and $U_{m+1} = 2tU_m - U_{m-1}$, and reduce modulo p and $U_{p-1}(x/2)$.

Now $\mathrm{FPdim}(L_2) = 2\cos(\pi/p) < 2$. Preservation of this dimension implies that $\mathrm{Fr}(L_2)$ is simple: every nonzero simple summand has Frobenius–Perron dimension at least one. Write it as $L_a \boxtimes L_b$. The identity of Lemma 19.5 says $b[L_a] = -2[L_{p-1}]$ in $K_0(\mathsf{Ver}_p) \otimes k$. Since $1 \le b < p$, comparison of basis coefficients gives $a = p - 1$ and $b = p - 2$. Finally apply Fr to $L_2 \otimes L_s = L_{s-1} \oplus L_{s+1}$ and use induction on s. The displayed fusion rules verify the two parity cases and give the formula. ∎

20.2 Frobenius type

20.2 Definition. The *Frobenius type* of $\mathcal{C}$ is the smallest fusion subcategory $\mathcal{A} \subseteq \mathsf{Ver}_p$ such that

$$\mathrm{Fr}(\mathcal{C}) \subseteq \mathcal{C}^{(1)} \boxtimes \mathcal{A}.$$

Equivalently, it is generated by all L_r occurring as a second factor in some simple summand of some $\mathrm{Fr}(X)$.

For $p > 3$, the possibilities are Vect, SVec, Ver_p^+, and Ver_p. To see why this list is exhaustive, a subcategory containing any L_r other than $\mathbf{1}$ or ε contains L_3 as a summand of $L_r \otimes L_r^*$. The fusion rules show that L_3 generates all the odd-indexed simples. Adding any even-indexed simple then generates ε and hence all of Ver_p.

20.3 Example. Both Vect and SVec have Frobenius type Vect; for the odd line π and odd p, the p-cycle acts trivially on $\pi^{\otimes p}$. Proposition 20.1 shows that Ver_p has type Ver_p^+. In particular the rank-two category Ver_5^+ has type Ver_5^+, as computed in the previous lecture. When $p = 2$, $\mathsf{Ver}_2 = $ Vect; when $p = 3$, $\mathsf{Ver}_3 = $ SVec and $\mathsf{Ver}_3^+ = $ Vect.

20.4 Proposition. *For an exact symmetric tensor functor $F : \mathcal{C} \to \mathcal{D}$, there is a natural monoidal isomorphism*

$$\mathrm{Fr}_{\mathcal{D}} \circ F \simeq (F^{(1)} \boxtimes \mathrm{id}) \circ \mathrm{Fr}_{\mathcal{C}}.$$

Consequently the Frobenius type of $\mathcal{C}$ is contained in that of $\mathcal{D}$.

Proof. The symmetric structure of F identifies the cyclic actions on $F(X^{\otimes p})$ and $F(X)^{\otimes p}$. In the semisimple construction, this identification is compatible with taking the multiplicity spaces and with the quotient of $\mathrm{Rep}_k(C_p)$ by negligible maps. It gives the stated isomorphism. Moreover an exact tensor functor is faithful, so it sends no nonzero object to zero. Thus every second-factor summand occurring before applying F still occurs afterwards; it must belong to the type of $\mathcal{D}$. ∎

In particular, a category admitting a super fiber functor has type Vect. If $\mathcal{C}$ itself has type Vect, its Frobenius functor lands in $\mathcal{C}^{(1)}$ and

$$[\mathrm{Fr}(X)] = [X]^p \quad \text{in } K_0(\mathcal{C}) \otimes_{\mathbb{Z}} k. \tag{20.1}$$

20.3 Scalar extension and split categories

For the discussion of lifting, temporarily let F be an arbitrary field. A finite semisimple F-linear category is *split* if $\mathrm{End}(S) = F$ for every simple object S. Equivalently, every simple remains simple after extension to an algebraic closure. Scalar extension is formed by extending morphism spaces and taking the additive idempotent completion; we write $\mathcal{C} \otimes_F F'$ for a field extension F'/F.

For example, the real two-dimensional representation of C_3 in which a generator acts by rotation through $2\pi/3$ is simple over $\mathbb{R}$. Its endomorphism algebra is $\mathbb{C}$, and after extension to $\mathbb{C}$ it splits into two one-dimensional characters. Thus $\mathrm{Rep}_{\mathbb{R}}(C_3)$ is not split.

20.4 The lifting theorem

For a symmetric fusion category, define its *global dimension* by

$$\dim(\mathcal{C}) = \sum_{S \text{ simple}} \dim(S)^2 \in k.$$

We call $\mathcal{C}$ *nondegenerate* when this scalar is nonzero. Here this word concerns global dimension; it does not mean that the braiding has trivial symmetric center.

Return to an algebraically closed field k of characteristic p, which is in particular perfect, and let $W(k)$ be its ring of Witt vectors. This is a complete discrete valuation ring of characteristic zero with residue field k. A split lifting of a split fusion category $\mathcal{C}$ to $W(k)$ consists of a rigid monoidal $W(k)$-linear category with finite free morphism modules, the same simple labels and fusion multiplicities, and reduction modulo p equivalent to $\mathcal{C}$. Extending to the fraction field gives a characteristic-zero fusion category.

20.5 Theorem (Etingof–Nikshych–Ostrik)**.** *A nondegenerate fusion category over an algebraically closed field k lifts to $W(k)$, uniquely up to equivalence with prescribed reduction. A braiding, and in particular a symmetry, lifts as well. Lifting is faithful on equivalence classes of nondegenerate categories.*

These are the lifting and faithfulness results of [ENO05, Section 9]. They are input theorems here; their proof uses vanishing of deformation cohomology. The completeness and residue-field conditions on the base ring are essential to this formulation: the theorem is not an assertion about every local ring with residue field k.

21 Lecture 21 (March 5, 2025, est.): Nondegeneracy and Frobenius injectivity

Scribe: Guillermo Sanmarco

Let k be algebraically closed of characteristic $p > 0$ and let $\mathcal{C}$ be a symmetric fusion category. We continue to use *nondegenerate* for $\dim(\mathcal{C}) \neq 0$.

21.1 Nondegenerate categories have super fiber functors

21.1 Proposition. *If $\mathcal{C}$ is nondegenerate, then it admits a symmetric tensor functor to* Vect *when $p = 2$, and to* SVec *when $p > 2$.*

Proof. Use Theorem 20.5 to lift $\mathcal{C}$ symmetrically over $W(k)$. Extend the generic fiber to an algebraic closure of $K = \operatorname{Frac} W(k)$. By Deligne's theorem for symmetric fusion categories in characteristic zero, the result is $\operatorname{Rep}_{\overline{K}}(G, z)$ for a finite group G and a central element z with $z^2 = 1$. Here z acts as parity and the symmetry is the super symmetry. Its global dimension is $|G|$: the superdimensions of simples differ from their ordinary dimensions only by signs, which disappear on squaring.

Global dimension of the integral lifting reduces to $\dim(\mathcal{C})$. Thus $|G|$ is not divisible by p. The category $\operatorname{Rep}_k(G, z)$ is consequently semisimple and nondegenerate, and its standard integral lifting has the same geometric generic fiber. Faithfulness of lifting, [ENO05, Corollary 9.9], identifies $\mathcal{C}$ with $\operatorname{Rep}_k(G, z)$. The use of the geometric generic fiber avoids assuming that the fraction field K is algebraically closed. For $p > 2$, take the forgetful functor to SVec. If $p = 2$, the group has odd order, so $z = 1$ and the forgetful functor lands in Vect. ∎

21.2 A dimension test for full faithfulness

For a tensor functor $F : \mathcal{C} \to \mathcal{D}$ between fusion categories, write $F(\mathcal{C})$ for its *fusion image*: the full subcategory closed under direct sums and direct summands generated by the objects $F(X)$. This need not be merely the list of objects literally equal to $F(X)$.

21.2 Lemma. *One has*

$$\operatorname{FPdim}(F(\mathcal{C})) \leq \operatorname{FPdim}(\mathcal{C}),$$

with equality if and only if F is fully faithful.

Proof. Replace the target by the fusion image $\mathcal{D}_0 = F(\mathcal{C})$; then F is dominant. Let R be its right adjoint. The standard dimension identity for a dominant tensor functor is

$$\operatorname{FPdim}(\mathcal{C}) = \operatorname{FPdim}(\mathcal{D}_0) \operatorname{FPdim}(R(\mathbf{1})).$$

For completeness, if N_{ia} is the multiplicity of the simple Y_a in $F(X_i)$, the vector $\sum_i \operatorname{FPdim}(X_i) N_{ia}$ is the image of the regular dimension vector of $\mathcal{C}$. It is therefore an eigenvector for multiplication by $F(\bigoplus_i X_i)$. Dominance means that this object contains every simple of $\mathcal{D}_0$; its multiplication matrix has strictly positive entries,

by rigidity. Perron–Frobenius uniqueness makes the displayed vector a multiple of $\mathrm{FPdim}(Y_a)$; its coefficient at the unit is $\mathrm{FPdim}(R(\mathbf{1}))$. Summing against $\mathrm{FPdim}(Y_a)$ gives the identity. The unit occurs once in $R(\mathbf{1})$ by adjunction, so this last dimension is at least one. Equality holds exactly when $R(\mathbf{1})$ is the unit. The projection formula $R(F(X)\otimes Y)\simeq X\otimes R(Y)$, obtained by rigidity and adjunction, then gives $RF\simeq \mathrm{id}$, which is equivalent to full faithfulness. Conversely a fully faithful F gives $R(\mathbf{1})\simeq\mathbf{1}$. ∎

21.3 Definition. The category $\mathcal{C}$ is *Frobenius injective* if $\mathrm{Fr}:\mathcal{C}\to\mathcal{C}^{(1)}\boxtimes \mathsf{Ver}_p$ is fully faithful.

The word "injective" in this definition means fully faithful, not merely faithful. Every tensor functor under consideration is already faithful.

21.3 A Grothendieck-ring criterion

21.4 Lemma. *If the finite-dimensional algebra $A=K_0(\mathcal{C})\otimes_{\mathbb{Z}} k$ is semisimple, then $\mathcal{C}$ is nondegenerate.*

Proof. Let $t(a)$ be the trace of left multiplication by a on A. Since A is commutative semisimple over the algebraically closed field k, it is a product of copies of k, and $(a,b)\mapsto t(ab)$ is nondegenerate. Set $r=\sum_Z[Z^*\otimes Z]$, summing over the simples. For simple X,Y, rigidity gives

$$t([X][Y])=\sum_Z \dim_k \mathrm{Hom}(X\otimes Y\otimes Z,Z)=\dim_k\mathrm{Hom}(X\otimes Y,\bigoplus_Z Z^*\otimes Z),$$

with the integer on the right read in k. The matrix of multiplication by r therefore agrees with the matrix of this trace pairing after permuting the basis by duality. It is invertible, so r is a unit of A. Apply the categorical-dimension homomorphism $A\to k$: the image of r is $\sum_Z \dim(Z)^2=\dim(\mathcal{C})$, which must be nonzero. ∎

21.5 Lemma (Type Vect)**.** *A Frobenius-injective category of Frobenius type* Vect *is nondegenerate.*

Proof. Here $\mathrm{Fr}:\mathcal{C}\to\mathcal{C}^{(1)}$ is fully faithful. The two categories have the same Frobenius–Perron dimension, so this is an equivalence. Equation (20.1) shows that taking pth powers permutes the simple-object basis of A, with the Frobenius automorphism on scalar coefficients. Thus the map $a\mapsto a^p$ is bijective. It is Frobenius-semilinear, rather than k-linear, but bijectivity still follows because k is perfect. An element killed by some power p^m must therefore be zero. Hence A is reduced, and a finite-dimensional commutative reduced algebra over k is semisimple. Apply Lemma 21.4. ∎

21.6 Lemma (Type SVec)**.** *Suppose $p>2$. A Frobenius-injective category of Frobenius type* SVec *is nondegenerate.*

Proof. Full faithfulness makes the image of a simple object simple. Thus for every simple X, either $\mathrm{Fr}(X)=Y\boxtimes\mathbf{1}$ or $\mathrm{Fr}(X)=Y\boxtimes L_{p-1}$. These alternatives give a faithful $\mathbb{Z}/2$-grading $\mathcal{C}=\mathcal{C}_0\oplus\mathcal{C}_1$. The neutral component has Frobenius type

Vect, and its Frobenius functor is fully faithful by naturality. Lemma 21.5 gives $\dim(\mathcal{C}_0) \neq 0$.

Here is the dimension calculation for the grading. Put $r_i = \sum_{X \in \mathcal{C}_i} \dim(X)[X]$ in $K_0(\mathcal{C}) \otimes k$. For a simple $Y \in \mathcal{C}_1$, the fusion rules and rigidity imply $[Y]r_0 = \dim(Y)r_1$; this follows by comparing the coefficient of each simple, using $N_{Y,X}^{Z} = N_{Y^*,Z}^{X}$. Apply categorical dimension and cancel $\dim(Y) \neq 0$ to obtain $\dim(\mathcal{C}_0) = \dim(\mathcal{C}_1)$. Consequently

$$\dim(\mathcal{C}) = 2\dim(\mathcal{C}_0) \neq 0.$$

■

21.4 The other Frobenius types

21.7 Lemma. *Suppose $p > 3$ and $\mathcal{C}$ has Frobenius type Ver_p^+ or Ver_p. Then its Frobenius fusion image contains $\mathbf{1} \boxtimes \mathsf{Ver}_p^+$.*

Proof. Some $\mathrm{Fr}(X)$ contains a simple summand $Y \boxtimes L_r$ with $r \neq 1, p-1$. Its tensor product with its dual contains $(Y \otimes Y^*) \boxtimes (L_r \otimes L_r^*)$. The first factor contains the unit, and the fusion rules show that the second contains L_3. Thus the fusion image contains $\mathbf{1} \boxtimes L_3$, which generates $\mathbf{1} \boxtimes \mathsf{Ver}_p^+$. ■

This lemma does not require Frobenius injectivity. Full faithfulness will be used in the next lecture to lift individual summands back to $\mathcal{C}$ and complete the proof of Ostrik's theorem.

22 Lecture 22 (March 7, 2025, est.): Ostrik's theorem

Scribe: Guillermo Sanmarco

Let k be algebraically closed of characteristic $p > 0$. We finish the proof that every symmetric fusion category over k admits a symmetric tensor functor to Ver_p. The argument follows the Frobenius functor and its fusion image, using the lemmas of the preceding lecture.

22.1 Separating the two factors

Assume first that $p > 3$, that $\mathcal{C}$ is Frobenius injective, and that its Frobenius type is Ver_p^+ or Ver_p. Define $\mathcal{C}_s$ to be the full fusion subcategory whose simple objects satisfy

$$\mathrm{Fr}(X) = Y \boxtimes \mathbf{1} \quad \text{or} \quad \mathrm{Fr}(X) = Y \boxtimes \varepsilon, \qquad \varepsilon = L_{p-1}.$$

This is indeed a fusion subcategory: it is the inverse image of $\mathcal{C}^{(1)} \boxtimes \mathrm{SVec}$ under Fr. By naturality, its own Frobenius functor is the restriction of the one on $\mathcal{C}$. Thus it is Frobenius injective and has type Vect or SVec. Lemmas 21.5 and 21.6 imply

$$\dim(\mathcal{C}_s) \neq 0. \tag{22.1}$$

22.1 Lemma. *Under these assumptions,*

$$\mathrm{Fr}(\mathcal{C}) = \langle \mathrm{Fr}(\mathcal{C}_s), \mathbf{1} \boxtimes \mathsf{Ver}_p^+ \rangle \simeq \mathcal{C}_s \boxtimes \mathsf{Ver}_p^+$$

as symmetric fusion categories.

Proof. Both subcategories on the right belong to the Frobenius image, the second by Lemma 21.7. For the reverse inclusion take a simple $X \in \mathcal{C}$. Its image is simple, so write $\mathrm{Fr}(X) = T \boxtimes L_r$. Put

$$\delta_r = \begin{cases} \mathbf{1}, & r \text{ odd}, \\ \varepsilon, & r \text{ even}. \end{cases}$$

Then $L_r \otimes \delta_r$ belongs to Ver_p^+. Full faithfulness and semisimplicity imply that every object in the fusion image is the image of an object: a direct summand is specified by an idempotent, which lifts through a fully faithful functor and splits in the source. Therefore there is a simple $Y \in \mathcal{C}$ with

$$\mathrm{Fr}(Y) = \mathbf{1} \boxtimes (L_r \otimes \delta_r).$$

Now $\mathrm{Fr}(X \otimes Y)$ has the simple summand $T \boxtimes \delta_r$, because L_r is self-dual and $\mathbf{1}$ is a summand of $L_r \otimes L_r$. Lift this summand to a simple Z of $X \otimes Y$. It satisfies $\mathrm{Fr}(Z) = T \boxtimes \delta_r$, hence $Z \in \mathcal{C}_s$, and

$$\mathrm{Fr}(X) \simeq \mathrm{Fr}(Z) \otimes \big(\mathbf{1} \boxtimes (L_r \otimes \delta_r)\big).$$

This proves generation.

It remains to justify that generation is a Deligne product. The two subcategories have only the unit as a common simple: a simple in $\mathrm{Fr}(\mathcal{C}_s)$ has second factor $\mathbf{1}$ or ε, whereas a simple in $\mathbf{1} \boxtimes \mathsf{Ver}_p^+$ has first factor $\mathbf{1}$ and odd-indexed second factor. The only common possibility is the unit. In a symmetric fusion category, two full fusion subcategories $\mathcal{A}, \mathcal{B}$ with this property have fully faithful product functor $\mathcal{A} \boxtimes \mathcal{B} \to \langle \mathcal{A}, \mathcal{B} \rangle$. Indeed, for simple $A_i \in \mathcal{A}$ and $B_i \in \mathcal{B}$, rigidity identifies

$$\mathrm{Hom}(A_1 \otimes B_1, A_2 \otimes B_2) \simeq \mathrm{Hom}(A_2^* \otimes A_1, B_2 \otimes B_1^*).$$

Only the unit can contribute to this Hom-space; it contributes once exactly when $A_1 \simeq A_2$ and $B_1 \simeq B_2$. The product functor is therefore fully faithful and, by generation, essentially surjective. Apply this with $\mathcal{A} = \mathrm{Fr}(\mathcal{C}_s)$ and $\mathcal{B} = \mathbf{1} \boxtimes \mathsf{Ver}_p^+$, and use $\mathrm{Fr}(\mathcal{C}_s) \simeq \mathcal{C}_s$. ∎

22.2 The theorem

22.2 Theorem (Ostrik)**.** *Every symmetric fusion category $\mathcal{C}$ over k admits a k-linear exact symmetric monoidal functor*

$$\mathcal{C} \longrightarrow \mathsf{Ver}_p.$$

Proof. Suppose there is a counterexample and choose one of minimal Frobenius–Perron dimension. Such a minimum exists. To check this point, fix a bound M. A fusion category of dimension at most M has at most M simple objects, because every simple has Frobenius–Perron dimension at least one. Its simple dimensions are at most $\sqrt{M}$, and its fusion multiplicities satisfy $N_{ij}^r \leq d_i d_j / d_r \leq M$. There are therefore only finitely many possible based fusion rings below the bound, and hence finitely many possible Frobenius–Perron dimensions.

By Lemma 21.2, the fusion image of $\mathrm{Fr} : \mathcal{C} \to \mathcal{C}^{(1)} \boxtimes \mathsf{Ver}_p$ has dimension at most that of $\mathcal{C}$. If it had smaller dimension, minimality would give a symmetric tensor functor from that image to Ver_p, and composition would contradict the choice of $\mathcal{C}$. Consequently equality holds and Fr is fully faithful.

If its type is Vect, or SVec when $p > 2$, the lemmas of Lecture 21 make $\mathcal{C}$ nondegenerate. Proposition 21.1 then supplies a functor to Vect or SVec, followed by the standard inclusion in Ver_p. This is a contradiction. These cases include every possibility for $p = 2, 3$.

Thus $p > 3$, and the type must be Ver_p^+ or Ver_p. By (22.1), the category $\mathcal{C}_s$ has a super fiber functor. Compose

$$\mathcal{C} \xrightarrow{\mathrm{Fr}} \mathrm{Fr}(\mathcal{C}) \simeq \mathcal{C}_s \boxtimes \mathsf{Ver}_p^+ \longrightarrow \mathrm{SVec} \boxtimes \mathsf{Ver}_p^+ \simeq \mathsf{Ver}_p.$$

This is the required functor, contradicting the counterexample. Exactness is automatic for an additive functor with semisimple source. ∎

22.3 Remark. This proves the existence statement in [Ost15, Theorem 1.5]. It does not say that $\mathcal{C}$ itself embeds in Ver_p: the functor need not be full. Nor does the theorem assert that every positive-characteristic symmetric tensor category is semisimple. The generalization beyond fusion categories requires Frobenius exactness, which is the subject of the next lecture.

23 Lecture 23 (March 10, 2025, est.): Beyond fusion categories

Scribe: Guillermo Sanmarco

This final lecture describes the extension of the Frobenius method to nonsemisimple categories, and the role of higher Verlinde categories. Throughout, k is algebraically closed of characteristic $p > 0$, and $\mathcal{C}$ is a symmetric tensor category with finite-dimensional Hom-spaces and finite-length objects. Recall that *moderate growth* means: for every X there is a real constant d_X such that $\ell(X^{\otimes n}) \leq d_X^n$ for every $n \geq 1$. The results and conjectures below retain the distinction made in the course between proved structure theorems and the proposed higher theory.

23.1 Frobenius functors in the general case

For $G \leq S_n$, the symmetry gives a functor

$$P_G : \mathcal{C} \longrightarrow \mathrm{Rep}(G, \mathcal{C}), \qquad X \longmapsto X^{\otimes n}$$

with the restricted permutation action. There is a general construction [CEO23, Section 3.1]

$$\mathrm{Fr}_G : \mathcal{C} \longrightarrow \mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}. \tag{23.1}$$

Its second step applies semisimplification in the G-representation variable. When $\mathcal{C}$ is not semisimple, this needs care: $\mathrm{Rep}(G, \mathcal{C})$ is not generally $\mathcal{C} \boxtimes \mathrm{Rep}_k(G)$, and quotienting *all* negligible morphisms of $\mathrm{Rep}(G, \mathcal{C})$ would also lose information from $\mathcal{C}$.

One precise construction presents $\mathcal{C}$ as comodules of a cotriangular copseudo-Hopf algebra H. Such a presentation records the abelian category by a coalgebra and its tensor structure by compatible bicomodule data. Then

$$\mathrm{Rep}(G, \mathcal{C}) \simeq H\text{-Comod}(\mathrm{Rep}_k(G)).$$

Apply $\mathrm{Rep}_k(G) \to \overline{\mathrm{Rep}_k(G)}$ to the representation-variable data and the coactions. The result is $H\text{-Comod}(\overline{\mathrm{Rep}_k(G)}) \simeq \mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}$. This construction is independent of the chosen presentation, and is symmetric monoidal; these facts are part of the cited construction.

The functor (23.1) is additive precisely when a Sylow p-subgroup of G acts transitively on $\{1, \ldots, n\}$. In particular, additivity requires $n = p^j$. In this case it is p^j-semilinear and becomes k-linear after replacing the first factor of its target by $\mathcal{C}^{(j)}$, with scalar action $\lambda \cdot f = \lambda^{p^j} f$.

The principal examples are

$$\mathrm{Fr} = \mathrm{Fr}_{C_p} : \mathcal{C} \longrightarrow \mathcal{C}^{(1)} \boxtimes \mathsf{Ver}_p$$

and the *enhanced Frobenius functor*

$$\mathrm{Fr}^{\mathrm{en}} = \mathrm{Fr}_{S_p} : \mathcal{C} \longrightarrow \mathcal{C}^{(1)} \boxtimes \overline{\mathrm{Rep}_k(S_p)}.$$

Neither is assumed exact in a general category.

23.2 Invariants, coinvariants, and the functor Fr_+

For $m \geq 1$, write

$$\Gamma^m X = (X^{\otimes m})^{S_m}, \qquad S^m X = (X^{\otimes m})_{S_m}.$$

These are respectively the largest subobject on which all permutations act trivially and the largest quotient with trivial permutation action. The inclusion followed by the quotient gives a canonical map $\Gamma^m X \to S^m X$.

23.1 Definition. For $j \geq 1$, define

$$\mathrm{Fr}_+^{(j)}(X) = \mathrm{im}\big(\Gamma^{p^j} X \longrightarrow S^{p^j} X\big), \qquad \mathrm{Fr}_+ = \mathrm{Fr}_+^{(1)}.$$

These are p^j-semilinear functors. The functor Fr_+ is a direct summand of $\mathrm{Fr}^{\mathrm{en}}$, extracted by the unit multiplicity in $\overline{\mathrm{Rep}_k(S_p)}$. It has a canonical lax monoidal structure; its structural maps are not asserted to be isomorphisms.

23.2 Example. For an ordinary vector space V with basis $v_1, \ldots, v_d$, the image of $\Gamma^{p^j} V \to S^{p^j} V$ has basis $v_1^{p^j}, \ldots, v_d^{p^j}$. Indeed, the orbit sum of a mixed tensor maps to its orbit size times the corresponding monomial. The multinomial coefficient for a nonconstant word of length p^j is divisible by p, whereas constant words have orbit size one. Thus this functor recovers the usual Frobenius twist of a vector space, and similarly the Frobenius twist of a representation.

23.3 Theorem (Coulembier–Etingof–Ostrik)**.** *The following conditions are equivalent:*

1. Fr *is exact;*
2. $\mathrm{Fr}^{\mathrm{en}}$ *is exact;*
3. Fr_+ *is exact.*

A category satisfying them is called Frobenius exact*.*

See [CEO23, Proposition 3.4]. The equivalence is a theorem, not a formal consequence of additivity. In particular, semisimple categories are Frobenius exact, but nonsemisimple examples need not be.

23.4 Theorem (The Verlinde fiber-functor criterion)**.** *The category $\mathcal{C}$ is Frobenius exact and of moderate growth if and only if it admits an exact k-linear symmetric monoidal functor $\mathcal{C} \to \mathsf{Ver}_p$.*

This is the general structure theorem of [CEO23, Theorem 7.13]. The forward implication is a deep extension of the fusion-category argument. For the reverse implication, exact faithful functors preserve a strict composition series and hence cannot decrease length. Tensor powers in the fusion category Ver_p have exponentially bounded length, proving moderate growth. Naturality of the Frobenius construction and exact faithfulness of the fiber functor then detect exactness of Fr, since the target is semisimple.

23.5 Lemma (Iteration)**.** *If $\mathcal{C}$ is Frobenius exact, then*

$$\mathrm{Fr}_+^{(j)} \simeq (\mathrm{Fr}_+)^{\circ j} \qquad (j \geq 1).$$

This is [CEO23, Lemma 5.1]; the equality compares the same semilinearity on both sides. Its proof uses a faithful exact symmetric functor that splits all short exact sequences, and the comparison of invariants and coinvariants for iterated permutation groups. Without Frobenius exactness the assertion can fail. Thus the definition using S_{p^j}-invariants must not be replaced by iteration without this hypothesis.

23.3 Higher Verlinde categories and incompressibility

23.6 Definition. A symmetric tensor category $\mathcal{D}$ is *incompressible* if every tensor functor from $\mathcal{D}$ to another symmetric tensor category is an embedding of a tensor subcategory: it is fully faithful and its image is closed under subobjects and quotients.

Besides Vect, SVec for $p > 2$, and Ver_p, there are finite symmetric tensor categories Ver_{p^n} and $\mathsf{Ver}_{p^n}^+$ for $n \geq 1$, constructed from quotients of the category of tilting modules for SL_2 and their abelian envelopes [BEO20]. There are tensor embeddings

$$\mathsf{Ver}_p \subseteq \mathsf{Ver}_{p^2} \subseteq \mathsf{Ver}_{p^3} \subseteq \cdots .$$

These categories and their indicated subcategories are incompressible. For $n > 1$, Ver_{p^n} is not Frobenius exact. In particular, moderate growth alone does not imply the existence of a fiber functor to Ver_p.

23.7 Theorem (Coulembier–Etingof–Ostrik)**.** *Every symmetric tensor category of moderate growth admits a tensor functor to an incompressible symmetric tensor category.*

This existence theorem is proved in [CEO24]. It leaves the classification of possible incompressible targets as a separate problem. The conjecture discussed in the course is that every category of moderate growth fibers over $\mathsf{Ver}_{p^\infty} = \bigcup_n \mathsf{Ver}_{p^n}$; the known higher Verlinde categories and their tensor subcategories would then supply all the incompressible targets in this class. This is stated as a conjecture, not used as a theorem.

23.4 A proposed higher Frobenius construction

We conclude with the construction of Coulembier–Flake [CF24]. Fix $n \geq 1$ and $E = (C_p)^n$. Choose a faithful two-dimensional representation V of E, equivalently an embedding $E \hookrightarrow \mathrm{SL}_2(k)$. For example, choose $\mathbb{F}_p$-linearly independent elements $a_1, \ldots, a_n \in k$ and send the ith generator to $\left(\begin{smallmatrix} 1 & a_i \\ 0 & 1 \end{smallmatrix}\right)$. Let $\mathcal{D}$ be the full additive idempotent-complete monoidal subcategory generated by V in $\mathrm{Rep}_k(E)$.

There is a tensor ideal of morphisms $\mathcal{K}$ in $\mathcal{D}$ such that Ver_{p^n} is the monoidal abelian envelope of $\mathcal{D}/\mathcal{K}$. This is an actual construction theorem. For $n = 1$ it recovers the

quotient of $\mathrm{Rep}_k(C_p)$ by negligible maps. Together with the regular permutation embedding $E \hookrightarrow S_{p^n}$, it leads to candidate functors

$$\Phi_{\mathcal{C}}^{V} : \mathcal{C} \longrightarrow \mathcal{C}^{(n)} \boxtimes \mathsf{Ver}_{p^n},$$

based on the E-action on $X^{\otimes p^n}$. The exponent in the Frobenius scalar twist is p^n.

The category $\mathcal{D}$ has only finitely many indecomposables and has a smallest nonzero *thick tensor ideal of objects* $\mathcal{J}$. Here "thick" means closed under finite direct sums and direct summands; being a tensor ideal means closure under tensoring with every object of the ambient category. This ideal of objects is distinct from the ideal of morphisms $\mathcal{K}$ above.

The two conjectural steps discussed in the handwritten notes are:

1. The candidate functors $\Phi_{\mathcal{C}}^{V}$ admit compatible symmetric monoidal structures and intertwine tensor functors in $\mathcal{C}$, giving higher analogues of the ordinary Frobenius functor.
2. The full subcategory $\mathcal{J} \subseteq \mathcal{D} \subseteq \mathrm{Rep}_k(E)$ remains a thick tensor ideal in the entire category $\mathrm{Rep}_k(E)$, including closure under tensoring with modules outside $\mathcal{D}$.

Coulembier–Flake prove that the second statement implies the first. They establish special cases, but the general monoidality assertion is not a consequence merely of taking a tensor power. The longer-term aim is an exactness criterion for fibering over Ver_{p^n}, parallel to Theorem 23.4 when $n = 1$.

References

[BEO20] Dave Benson, Pavel Etingof, and Victor Ostrik. New incompressible symmetric tensor categories in positive characteristic, 2020.

[CEO23] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. On Frobenius exact symmetric tensor categories. *Annals of Mathematics*, 197(3):1235–1279, 2023.

[CEO24] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. Incompressible tensor categories. *Advances in Mathematics*, 457:109935, 2024.

[CF24] Kevin Coulembier and Johannes Flake. Towards higher Frobenius functors for symmetric tensor categories, 2024. The course discussion follows the May 2024 version.

[Del02] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[DM82] Pierre Deligne and James S. Milne. Tannakian categories. In *Hodge Cycles, Motives, and Shimura Varieties*, volume 900 of *Lecture Notes in Mathematics*, pages 101–228. Springer, 1982. Corrected version, 2022, with the original proposition numbering.

[EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.

[ENO05] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. On fusion categories. *Annals of Mathematics*, 162(2):581–642, 2005.

[EO19] Pavel Etingof and Victor Ostrik. On semisimplification of tensor categories, 2019.

[Mil12] James S. Milne. Basic theory of affine group schemes, 2012. Available at www.jmilne.org/math/.

[Ost15] Victor Ostrik. On symmetric fusion categories in positive characteristic, 2015.

[Wat79] William. C. Waterhouse. *Introduction to Affine Group Schemes*. Springer Graduate texts in mathematics. Springer New York, NY, 1979.

[Wib25] Michael Wibmer. A remark on torsors under affine group schemes. *Transformation Groups*, 30:447–454, 2025.